Dell CloudLink 8.3.x

Deployment Guide

June 2026
Rev. 1.0

DELL Technologies

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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Revision History

Table 1. Revisions

Date	Document Revision	Description of Changes
June 2026	1.0	Initial release

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About Dell CloudLink

About Dell CloudLink for PowerFlex

Enterprises have many reasons for encrypting their data—addressing regulatory compliance, protecting against theft of customer data, and sensitive intellectual property.

CloudLink offers significant benefits for environments that use Dell PowerFlex resources. PowerFlex is a software-defined solution that enables you to transform Direct Attached Storage (DAS) on existing hardware into shared block storage. It offers considerable scalability and extreme performance with flexible and elastic storage capacity and nodes.

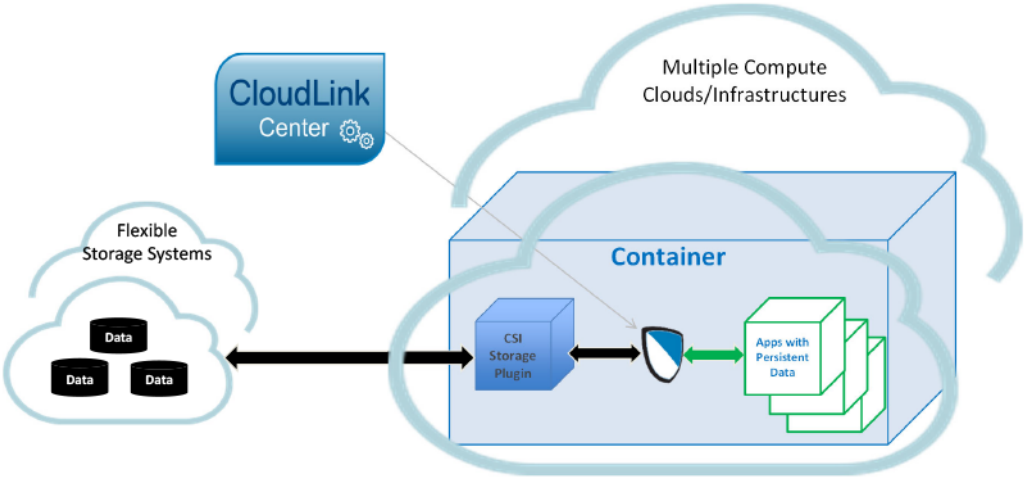
CloudLink provides software-based Data at Rest Encryption (DARE) for PowerFlex storage node that is transparent to the features and operation of the PowerFlex solution. It uses dm-crypt, a native Linux encryption package, to secure storage node devices. A proven high-performance volume encryption solution, dm-crypt is widely implemented for Linux machines.

CloudLink encrypts the storage node devices with unique keys which the enterprise security administrators control. CloudLink Center provides centralized, policy-based management for these keys, enabling single-screen security monitoring and management across one or more PowerFlex deployments.

About Dell CloudLink for Containers

CloudLink supports data encryption in a Kubernetes containerized environment. CloudLink encryption for containers enables you to encrypt shared volumes in a Kubernetes cluster. This functionality leverages Kubernetes 1.14 to 1.21 Container Storage Interface (CSI), which is customizable to the user environment, and features a quick, easy setup with the UI or REST-API.

Encryption of Containers Agents sits between the Application and the CSI Storage Plugin encrypting the application data before it is sent to storage-thus providing both Data at Rest and Data in Motion. One CloudLink Center instance can support multiple Kubernetes clusters. Each Kubernetes cluster node can have multiple Container agents running on it, which includes one Encryption for Containers agent for each driver.



About Dell CloudLink Deployment Guide

This guide provides step-by-step instructions for installing, deploying, and configuring Dell CloudLink.

Intended audience for the CloudLink Deployment Guide

The CloudLink deployment guide is intended for IT administrators who are responsible for deploying and maintaining machines in the CloudLink Center environment, but not necessarily for the security of data on those machines.

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CloudLink 8.3.x license deployment or migration offerings

If you are upgrading to CloudLink 8.3.x from an earlier version, the license of the earlier version is retained and is displayed on the CloudLink interface. However, for 8.3.x and later, you do not require explicit licenses to access CloudLink.

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Overview of CloudLink Center deployment scenarios

CloudLink components

Dell CloudLink consists of the following components.

- CloudLink Center—The web-based interface for CloudLink that is used to manage machines that belong to the CloudLink environment (those machines on which CloudLink Agent is installed).

  CloudLink Center:

  - Communicates with machines over Transport Layer Security (TLS)
  - Manages the encryption keys that are used to secure the boot volumes, data volumes, and devices for the machines
  - Configures the security policies
  - Monitors the security and operation events
  - Collects log data
- CloudLink Agent—The agent that runs on individual machines. It communicates with CloudLink Center for prestartup authorization and decryption of BitLocker or dm-crypt encryption keys.

NOTE: You can install an agent on a physical server. However, CloudLink cannot be deployed on a physical server.

For Enterprise and PowerFlex—CloudLink Center is packaged as a virtual appliance that can be deployed in the enterprise on VMware ESXi or Microsoft Hyper-V. Download CloudLink Agent from CloudLink Center.

For Microsoft Azure or Azure Stack—CloudLink Center can be deployed from the Azure Gallery in a simple-to-deploy, self-contained image file that enables you to quickly start your business-critical operations by using CloudLink. Search the Azure Gallery for CloudLink to locate the image. Download CloudLink Agent from CloudLink Center.

CloudLink Center deployment model for Enterprise

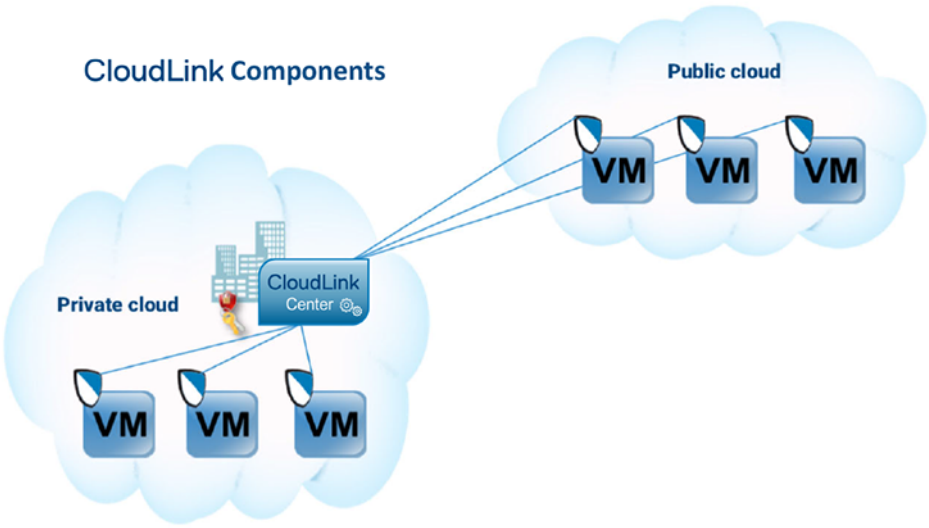
This guide assumes that CloudLink Center and the encryption keystore are deployed in the private cloud, as shown in the following figure. CloudLink Agent is deployed to individual VMs hosted in the private cloud or to VM instances in a supported public or hybrid cloud environment.

NOTE: CloudLink is a (closed) virtual appliance that does not support installation of third party applications.

When deployed, CloudLink Agent replicates the machine networking configuration as required, to ensure that it can communicate with CloudLink Center during the startup process. This replication to the preboot environment includes the IP configuration for available network interfaces and any static routing information. For Linux machines, if the networking configuration is changed after deployment, you must refresh the CloudLink Agent service. For more information about CloudLink Agent service, see Refresh the CloudLink Agent service on Linux machines in Custom mode.

NOTE: Dell Technologies recommends that you enable EFI boot mode of all CloudLink agent-installed VMs for enhanced security against boot-level attacks. However, the following are not supported:

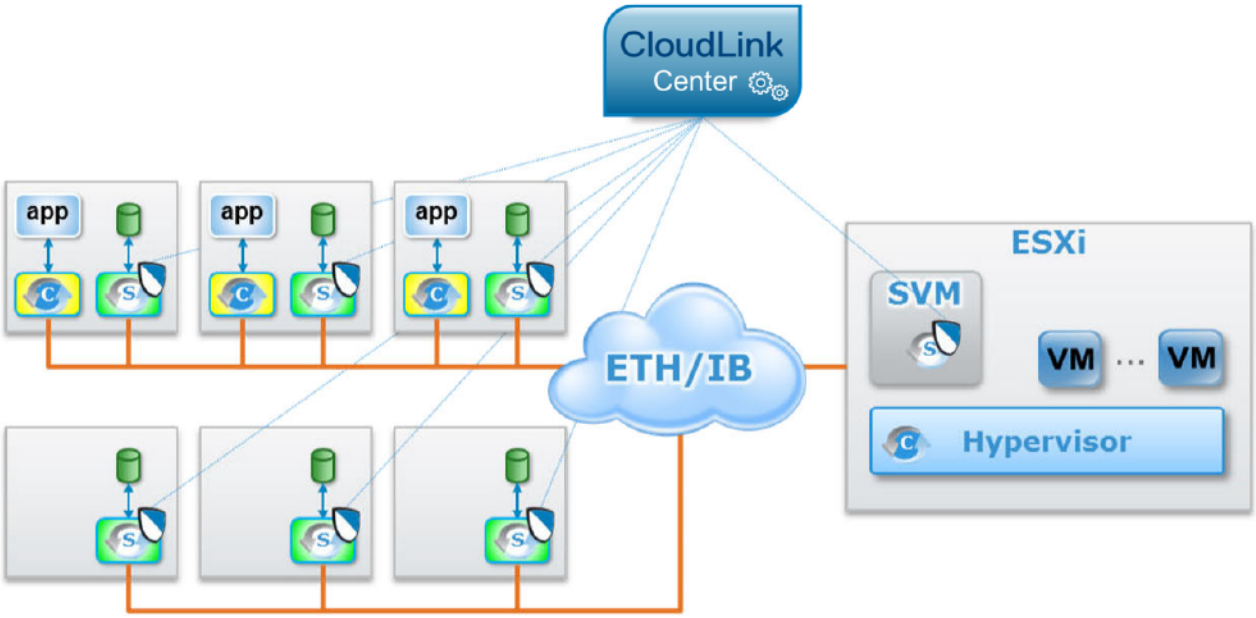
- The Secure Boot feature
- IPv6 version with EFI



CloudLink Center deployment model for PowerFlex

CloudLink supports various PowerFlex environments by deploying agents on physical and virtual Linux storage node devices, accommodating fully converged, two-layer, and mixed configurations.

CloudLink can be deployed to support various PowerFlex environments, as illustrated in the following figure. The CloudLink Agent can be deployed on physical and virtual Linux PowerFlex storage nodes and supports fully converged, two-layer, and mixed configurations. CloudLink Center must be accessible by all encrypted storage nodes.



Verify the ISO signature file used in CloudLink Center

To ensure that the ISO file you use to upgrade CloudLink is valid and secure, do the following:

Prerequisites

Ensure that OpenSSL is installed on the system where CloudLink Center is deployed.

Steps

1. Download the public key (PEM), binary signature file of the certificate (BIN), and the ISO file to a directory.
2. From the directory where the aforesaid files are saved, run the following command at the OpenSSL interface: `openssl dgst -sha256 -verify publickey.pem -signature clc-8.0.1-131.81.cert.bin clc-8.0.1-131.81.iso`.
   The file is verified and the following message is displayed: **Verified OK**.

Software requirements for deploying CloudLink Center

This section describes the system requirements for CloudLink Center, and the machines on which CloudLink Agent will be deployed. Ensure that these requirements are met before deployment.

System requirements for deploying CloudLink Center in Enterprise and PowerFlex

- For VMware deployments:
  - vSphere 6.0 or later
- For Microsoft Hyper-V deployments:
  - Hyper-V for Windows Server 2008 R2, Windows Server 2012 R2, Windows Server 2016, or Windows Server 2019
- VMware Virtual Volume (vVol) and VMware vStorage APIs for Storage Awareness (VASA):
  - vVol 1.0, VASA 2.0: vSphere 6.0 or later
  - vVol 2.0, VASA 3.0: vSphere 6.5 or later
  - vVol 1.0 and VASA 2.0 on PowerFlex 3.5
  - vVol 2.0 and VASA 3.0 REST on PowerStore 1.0.3.1.3
- 4 vCPUs (minimum)
- 6 GB vRAM (minimum)
- 64 GB disk space
- Python package
- Web browser—Google Chrome 38 or higher or Mozilla Firefox 28 or higher

  TLS 1.2 must be enabled in your browser settings to connect to CloudLink Center. By default, CloudLink Center is enabled on browsers such as Mozilla Firefox (58 and later versions) and Google Chrome (64 and later versions).

CloudLink virtual appliance host machine requirements

For information about supported platforms, see the Dell CloudLink 8.3.x Release Notes.

Network port information for CloudLink Center in PowerFlex

This topic provides network port information for CloudLink Center in PowerFlex.

Table 2. CloudLink network ports

Port	TCP	UDP	Service
Incoming			
4443	Yes	No	CloudLink cluster communication
443	Yes	No	CloudLink Center web access, cluster communication, and Agent download.
1194	Yes	Yes	CloudLink Agent communication
5696	Yes	No	Key Management over KMIP

Table 2. CloudLink network ports (continued)

Port	TCP	UDP	Service
50000	Yes	No	Encryption for Containers
Outgoing			
123	No	Yes	Network Time Protocol (NTP)
443	Yes	No	External keystores other than Microsoft Active Directory
514	No	Yes	Syslog

Supported maximum CloudLink and KMIP values

CloudLink Center, when configured with 4 vCPUS, 8 GB RAM, and 32 GB disk space, can support the following maximum values:

- CloudLink Agents: 2,500
- CloudLink Agent encryption keys: 1 Million keys in SLES 8.0
- KMIP partitions: 100
- KMIP encryption keys (total): 50,000
- CloudLink cluster nodes: 4
- Virtual machine groups: 100

NOTE: A single CloudLink cluster supports the maximum values for either CloudLink Agent-based encryption or KMIP, but not both.

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Deploy and configure CloudLink Center

This chapter provides instructions for deploying and configuring CloudLink Center on:

- VMware vSphere
- Microsoft Hyper-V
- Microsoft Azure
- Microsoft Azure Stack
- Linux KVM

NOTE: By default, CloudLink Center is deployed in the self-certificate-signed mode. To change it to the third-party-signed mode, see the "Secure CloudLink Center agents using mutual authentication" chapter in the *Dell CloudLink Administration Guide* available on the support site.

Connect to the CloudLink Center console

Use this procedure to connect to the CloudLink Center console.

Steps

1. Do one of the following:
   - In vSphere Client, right-click the CloudLink Center virtual machine and select **Open Console**.
   - In Hyper-V Manager, right-click the CloudLink Center virtual machine and click **Connect**.
   - Start an SSH session to the CloudLink Center machine.
2. The default console login name is **`cloudlink`**.

   You changed the default console password the first time you logged in to the console or CloudLink Center.

Deploy and configure CloudLink Center in VMware vSphere for Enterprise and PowerFlex

CloudLink uses an interface to enable CloudLink Center to communicate with the CloudLink Agent that is installed on individual machines. This interface is supported through a virtual network interface that is in the Open Virtualization Format (OVF) template that is used to deploy CloudLink Center. The interface is configured when you first log in to CloudLink Center.

Deploy the CloudLink Center OVF template

Use this procedure to deploy the OVF template.

Prerequisites

This procedure assumes that you have obtained the CloudLink Center Open Virtualization Format (OVF) template that is used for deployment.

Steps

1. From vSphere Client, select **File** > **Deploy OVF Template**.
2. In the **Deploy OVF Template** window, go to your template folder, select the CloudLink Center template, and then click **Next**.
3. Ensure that the OVF template information is correct and click **Next**.

4. Enter a name, select an inventory location for the deployed template, and click **Next**.
5. Select a host or cluster to run the deployed template, and then click **Next**.
6. If a series of warnings is displayed, click **Yes** to continue with the deployment.
7. Select a resource pool and click **Next**.
8. In the **Review details** panel, verify the template details, and then click **Next**.
9. Select a storage location for the machine files and click **Next**.
10. Select the disk format for the virtual disk and click **Next**.
11. Select a destination network and click **Next**.
12. In the **Deployment Settings** panel, review the selected options and click **Finish**.
13. In the **Deployment Completed Successfully** dialog box, click **Close**.
    The CloudLink Center VM is displayed in the VMware vSphere VM list.
14. In vSphere Client, right-click the CloudLink Center VM and select **Power** > **Power on**.

Results

The DHCP server is assigned as the default server for CloudLink Center. If a DHCP server is available on your network, the CloudLink Center VM automatically generates a random hostname for the CloudLink Center by using the DHCP server. You can also preconfigure static network settings of CloudLink VM during deployment by using the vSphere vApp Configure option. When the static information is not provided during deployment, by default, the network configuration settings are assigned by a DHCP server. See the *Pre-configure the static network settings during CloudLink VM deployment* technical white paper available on the support site.

If a DHCP server is available, see Access CloudLink Center.

To configure static IPv4 or IPv6 network values, see Configure static network values in vSphere Client, Hyper-V Manager or KVM.

To configure DHCP IPv4 or IPv6 network values, see Configure DHCP IPv4 or IPv6 network values in vSphere Client or Hyper-V Manager or Virtual Machine Manager.

NOTE: Before you log in to CloudLink Center or the console, wait until vSphere Client reports that VMware Tools are installed and running. Else, you might experience network configuration issues.

After deploying OVF or VHD template or qcow2 image file

This topic provides information about the rules that you should consider after deploying an OVF or VHD template or qcow2 image file, and if you use IPv6 on your network.

NOTE: If you use IPv6 on your network, the following rules apply to CloudLink Center initial configuration:

- CloudLink supports the following combinations of IPv4 and IPv6 with static and dynamic network values:
  - IPv4 static
  - IPv4 DHCP
  - IPv6 static
  - IPv6 DHCP
- CloudLink Center or a CloudLink Center cluster can use IPv6 addresses (static, link-local, Stateless Address Autoconfiguration, or DHCPv6) to communicate with machines.
- When powered on, the CloudLink Center appliance tries to detect a DHCPv4 server or a DHCPv6 server. If not detected, the Initial Configuration wizard is automatically closed. By clicking Network in the Update menu, configure the IPv4 or IPv6 static values. See Configure static network values in vSphere Client, Hyper-V Manager or KVM.

Guidelines for configuring IPv4 or IPv6 addresses

Consider the following guidelines for configuring IPv4 or IPv6 or both IPv4 and IPv6 addresses.

- You can configure IPv4 or IPv6 or both IPv4 and IPv6 addresses on all the interfaces.
- You can choose not to configure IPv4 or IPv6 addresses on all the interfaces.
- You can add manual DNS entries when IPv4 or IPv6 or both IPv4 and IPv6 static network values are configured on at least one interface.
- You cannot add manual DNS entries in the DHCP mode.

- IPv6 network configuration is disabled after the network is reset.
  - When the system is restarted for the first time, if DHCPv6 is available in the network, IPv6 DHCP is assigned by default to the network. IPv6 network configuration is disabled after the network is reset.
  - IPv6 is enabled only after IPv6 static or DHCP address is set.
  - If only IPv4 is configured, IPv6 will be disabled. Reset the network and configure IPv6 again.
- DNS configuration and network configuration information is lost when the network is reset. If you exit the CloudLink Center console without configuring either IPv4 or IPv6 addresses, previous network configuration is not restored.
- Network configuration information is not restored when CloudLink Center is restored from a backup file.
- If your environment consists of a mixed configuration, you can choose to configure both IPv4 and IPv6 addresses for the CloudLink Center.
- If you add a secondary network interface after the system restarts, then the secondary network interface will not be configured by default. Reset the network to configure the secondary network interface.

Configure static network values in vSphere Client or Hyper-V Manager or Virtual Machine Manager

The following procedure explains how to configure IPv4 or IPv6 or both IPv4 and IPv6 static network values in vSphere Client, Hyper-V Manager, or Virtual Machine Manager.

Prerequisites

If a DHCP server is not available, set static IPv4 or IPv6 network values. Use the CloudLink Center console to configure the network settings.

NOTE: The static IPv6 network values are not supported in CloudLink 7.0.1 and earlier versions.

Steps

1. To configure static network values:
   - In vSphere Client or Virtual Machine Manager, right-click the CloudLink Center virtual machine and select **Open Console**.
   - In Hyper-V Manager, right-click the CloudLink Center virtual machine and click **Connect**.

     If the screen is blank, press Alt+F1 to open a new console window.
2. Log in to the CloudLink Center console with the login name **`cloudlink`** and the default password **`cloudlink`**.

   Change the default password.
3. When prompted, type a new password for the CloudLink Center console.
4. Press **down arrow** and type the password again to confirm it.
5. Press **Tab** and then **OK** to accept the password change.

   Subsequent logins to the console prompt for the new password, which you can change at any time from the Update Menu in the CloudLink Center console. For more information, see CloudLink Update Menu.
6. Press **OK** again in the **Summary** screen.
7. Select **Reset Network** in the **Update Menu**.
8. When prompted **Are you sure you want to reset all network configuration?**, select **Yes** and then press **Enter**.
9. In the **Network 1 Configuration** window, select either IPv4 or IPv6 based on your requirement, and then press **OK**.
10. Select **Static**, and then press **Enter**.
11. Type the IP address, netmask, and gateway address for CloudLink Center.
12. Press **Tab** and then **OK**.
    The **Network 1 Configuration** window is displayed.
13. Repeat steps 9 to 12 if you want to configure both IPv4 and IPv6.

    For example if you have:
    - Completed configuring IPv4 and want to configure IPv6, then select **IPv6** and repeat steps 9 to12.
    - Completed configuring IPv6 and want to configure IPv4, then select **IPv4** and repeat steps 9 to12.
    - Completed configuring IPv6 and want to configure IPv4, then select **IPv4** and repeat steps 9 to12.
14. Press Exit if you do not want to configure IPv4 and IPv6.

NOTE: If you reset the network and exit without configuring IPv4 or IPv6, CloudLink Center remains unconfigured.

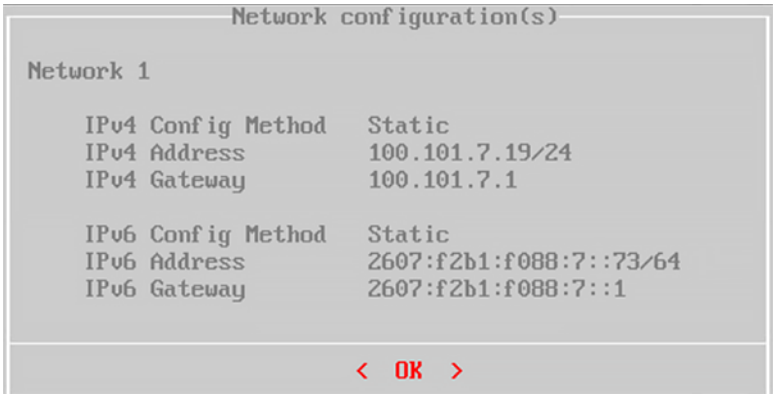
15. Wait for the network configuration to complete.

    This process might take some time.

16. If you want to configure multiple interfaces, repeat steps 9 to 15.

Results

After CloudLink Center network configuration is complete, a summary of its network settings is displayed. These settings include the IPv4 address or IPv6 address or both IPv4, and IPv6 addresses used to access CloudLink Center from a web browser and network configuration information.



Press **OK** to close the **Network configuration(s)** screen and return to the Update Menu. You can log out of the console.

The Update Menu is displayed every time you log in to the CloudLink Center console. For more information about using the Update Menu in CloudLink Center console, see CloudLink Update Menu.

Configure DHCP IPv4 or IPv6 network values in vSphere Client or Hyper-V Manager or Virtual Machine Manager

The following procedure explains how to configure IPv4 or IPv6 or both IPv4 and IPv6 DHCP network values in vSphere Client, Hyper-V Manager, or Virtual Machine Manager.

Prerequisites

If a DHCP server is not available, set static IPv4 or IPv6 network values. Use the CloudLink Center console to configure the network settings. For more information about configuring static network values, see Configure static network values in vSphere Client or Hyper-V Manager or Virtual Machine Manager.

Steps

1. To configure DHCP network values:
   - In vSphere Client or Virtual Machine Manager, right-click the CloudLink Center virtual machine and select **Open Console**.
   - In Hyper-V Manager, right-click the CloudLink Center virtual machine and click **Connect**.

     If the screen is blank, press Alt+F1 to open a new console window.
2. Log in to the CloudLink Center console with the login name **`cloudlink`** and the default password **`cloudlink`**.

   Change the default password.
3. When prompted, type a new password for the CloudLink Center console.
4. Press **down arrow** and type the password again to confirm it.
5. Press **Tab** and then **OK** to accept the password change.

   Subsequent logins to the console prompt for the new password, which you can change at any time from the Update Menu in the CloudLink Center console. For more information, see CloudLink Update Menu.
6. Press **OK** again in the **Summary** screen.

7. Select **Reset Network** in the **Update Menu**.
8. When prompted **Are you sure you want to reset all network configuration?**, select **Yes** and then press **Enter**.
9. In the **Network 1 Configuration** window, select either IPv4 or IPv6 based on your requirement, and then press **OK**.
10. Select **DHCP**, and then press **Enter**.
    Based on your selection, DHCP address is configured for either IPv4 or IPv6.
11. Repeat steps 9 to 12 if you want to configure both IPv4 and IPv6.

    For example if you have:
    - Completed configuring IPv4 and want to configure IPv6, then select **IPv6** and repeat steps 9 to12.
    - Completed configuring IPv6 and want to configure IPv4, then select **IPv4** and repeat steps 9 to12.
12. Press Exit if you do not want to configure IPv4 and IPv6.

    NOTE: If you reset the network and exit without configuring IPv4 or IPv6, CloudLink Center remains unconfigured.
13. Wait for the network configuration to complete.

    This process might take some time.
14. If you want to configure multiple interfaces, repeat steps 9 to 14.

Results

After CloudLink Center network configuration is complete, a summary of its network settings is displayed. These settings include the IPv4 address or IPv6 address or both IPv4, and IPv6 addresses used to access CloudLink Center from a web browser and network configuration information.

Press **OK** to close the **Network configuration(s)** screen and return to the Update Menu. You can log out of the console.

The Update Menu is displayed every time you log in to the CloudLink Center console. For more information about using the Update Menu in CloudLink Center console, see CloudLink Update Menu.

Deploy and configure CloudLink Center in KVM

CloudLink uses an interface to enable CloudLink Center to communicate with the CloudLink Agent that is installed on individual machines. This interface is supported through a virtual network interface that is included in the qcow2 image file which is used to deploy CloudLink Center. The interface is configured when you first log in to CloudLink Center.

Configure CloudLink static IP network in KVM

You can configure the CloudLink static IP network in KVM.

Prerequisites

Ensure that the latest version of the Linux Guestfish tool is installed on the virtual machine where you want to configure the static IP network.

About this task

To configure the static IP network, perform the following steps:

Steps

1. Copy the files, `Cloudlink-Build.qcow2` and `ifcfg-eth0`, to the KVM server.
2. In the Linux terminal, run the command:

```
virt-install --name VmDisplayName --ram 6122 --vcpus=4 --disk Cloudlink-
Build.qcow2,bus=sata --network bridge=BridgeNetwork,model=virtio --import --
noautoconsole --noreboot
```

3. Using the Guestfish tool, copy the network files to the `mountpoint` folder.

```
guestfish add Cloudlink-Build.qcow2 : run : mount /dev/sda10 / : copy-in ifcfg-eth0
/etc/sysconfig/network/
```

4. Start the virtual machine using the command:

```
virsh start VmDisplayName
```

To configure an IPv4 static address, use the parameter `IPADDR_1`, in the network configuration file. To configure an IPv6 static address, use `IPADDR_0`.

For example:

```
BOOTPROTO='static'
IPADDR_0='fe80::5054:ff:fef1:fe76/64'
IPADDR_1='100.101.7.57'
MTU='1500'
NAME='eth0'
NETMASK_1='255.255.255.0'
STARTMODE='auto'
USERCONTROL='no'
ONBOOT='yes'
```

Deploy the CloudLink Center qcow2 image file

The following procedure explains how to deploy the qcow2 image file.

Prerequisites

Ensure that you have obtained the CloudLink Center qcow2 image file.

Steps

1. In the Linux desktop environment, start **Virtual Machine Manager**.
2. Click **File** > **New Virtual Machine**.
   The **Create a new virtual machine** window appears.
3. Select **Import existing disk image**, and then click **Forward**.
4. Click **Browse** to select the CloudLink Center qcow2 image file, and then click **Forward**.
5. In the **Choose Memory and CPU settings** window, enter the following values:
   a. In the **Memory** box, enter 6 GB (6144 MB in binary).
   b. In the **CPU** box, enter 4 CPUs.
6. Click **Forward**.
   The **Ready to begin the installation** window appears.
7. In the **Name** box, enter a name for the VM.
8. Select **Customize configuration before install**.
9. From the **Network selection** list, select the destination network interface.
10. Click **Finish**.
    The **Customization** dialog appears.
11. Select **VirtIO Disk1** on the left pane.
    The **Virtual Disk** window appears.
12. Select SATA in the **Disk bus** box and click **Apply**.

Results

CloudLink Center assigns the DHCP server as the default server. If a DHCP server is available on your network, the CloudLink Center VM automatically generates a random hostname for the CloudLink Center by using the DHCP server.

If a DHCP server is available, see Access CloudLink Center.

To configure static IPv4 or IPv6 network values, see Configure static network values in vSphere Client, Hyper-V Manager or KVM.

To configure DHCP IPv4 or IPv6 network values, see Configure DHCP IPv4 or IPv6 network values in vSphere Client or Hyper-V Manager or Virtual Machine Manager.

Next steps

If you use IPv6 on your network, see After deploying OVF or VHD template or qcow2 image file.

Deploy and configure CloudLink Center image in Amazon Web Services (AWS)

AWS CloudLink Center image is required to deploy and configure CloudLink Center in AWS. Contact Dell Technologies Customer Support team to obtain AWS CloudLink Center image.

NOTE: CloudLink 8.0.x and 7.0.x images are currently unavailable in AWS platform . For information about the CloudLink image availability, contact Dell Technologies Support team.

NOTE: While deploying and configuring CloudLink Center in AWS, you are prompted to enter or accept the CloudLink hostname. This CloudLink hostname changes each time the CloudLink Center restarts.

Prerequisites to install VASA provider on controller nodes using PowerFlex

- SVM template with OpenJDK version 8 for VMware vSphere API for Storage Awareness (VASA) VMs
- VASA VM should have sufficient memory. The minimum required amount is 8 GB, though 16 GB is recommended.
- VASA VM with three network interface controllers for ESXi management, DATA 1, and DATA 2 networks
- The DNS server must be configured.
- The hostname of each VASA VM must match the VASA FQDN.
- OpenJDK version 8 must be installed. The `jdk-8u121-linux-x64.rpm` file is downloaded to `/root/install` directory.
- All components of your environment, including ESXi and vCenter, must have their time synchronized.
- A test folder is created and LIA, SDC, and VASA provider RPMs are copied.

6

Prepare CloudLink Agent for deployment

After deploying and configuring CloudLink Center, prepare to instal CloudLink agent to machines by accessing CloudLink Center and setting up CloudLink licenses.

Access CloudLink Center

This topic provides information about how to access CloudLink Center.

Ensure that the following prerequisites are met:

- Web browser to log in to CloudLink Center after it is deployed.
- HTTPS and JavaScript must be enabled in the web browser.
- URL or `clc_address` of the CloudLink Center

If you have a DHCP server on your network, the `clc_address` is available in the virtual machine of the CloudLink Center. To locate the clc_address:

- In vCenter Client, click **General panel**.
- In Hyper-V Manager, click **Networking**.

If you configured network settings in the CloudLink Center console, the `clc_address` is available in the **Update Menu** > **Summary** of the CloudLink Center console.

It is recommended that you configure Microsoft Windows domain integration so that you can access CloudLink Center with Windows domain credentials. In this case, do not provide CloudLink Center credentials. The secadmin user account remains a local account. For information about user accounts and configuring Microsoft Windows domain integration, see the "Microsoft Windows domain for user accounts" section in the *Dell CloudLink Administration Guide*.

It is recommended that you integrate to AD/LDAP with TLS, and integrating AD/LDAP with SSL is not recommended.

NOTE: CloudLink supports Azure Premium SSD v2 type of disks at configuration of 512b logical sector size only.

Log in to CloudLink Center by using web browser

Use this procedure to log in to CloudLink Center by using web browser.

Steps

1. In your web browser, type the URL for CloudLink Center in the following format: https://*clc_address*.
   To locate the *clc_address*, see Access CloudLink Center.
2. Log in with the username **`secadmin`**, and the secadmin password specified during deployment.

Prerequisites for configuring a CloudLink server

Ensure that the following prerequisites are met before configuring a CloudLink server.

- CloudLink license files determine the number of machine instances or the encrypted storage capacity that your organization can manage with CloudLink Center, and the duration of the license. For information about CloudLink license files, see the *Dell CloudLink Administration Guide*.
- The cluster server name is used primarily for CloudLink Center clusters, but it must be specified even if you do not plan to use clusters. See Requirements for CloudLink Center server addresses in clusters.
- Backups of CloudLink Center are encrypted using AES-256 with an RSA-2048 key pair. CloudLink Center only stores the public key. The private key must be downloaded and saved securely. Use CloudLink Web UI to generate an RSA-2048 key

pair. Ensure that you save the private key in a known and secure location. The private key is required when restoring a backup.

- CloudLink Center stores sensitive information that is encrypted in the CloudLink Vault. The CloudLink Vault has to be unlocked before the information can be accessed. It is recommended that you use the **Manual Unlock** mode when deciding about how CloudLink Center opens the CloudLink Vault when restarted. This mode opens CloudLink Vault only when an administrator provides an appropriate passcode.

  To ensure that CloudLink Vault can be unlocked manually after a CloudLink Center restart, use CloudLink Web UI to configure at least one CloudLink Vault passcode. See CloudLink Vault to encrypt CloudLink Center machine data.

Configure the CloudLink Center server

Use this procedure to access CloudLink Center, and configure the server. For Azure Stack, use this procedure only if you selected the **No Configuration** deployment type.

Steps

1. In your web browser, type the URL for CloudLink Center in the following format: https://*clc_address*.

   The *clc_address* must be in either FQDN, IPv4, or IPv6 format. To locate the *clc_address*, see Access CloudLink Center.
2. Accept the license agreement.
3. The password for the in-built secadmin user must be changed the first time you log in. Type the secadmin password, reenter it to confirm, and then click **Change Password**.

   You can change the password for the secadmin user account at any time after the first-time login. For more information, see the *Dell CloudLink Administration Guide*.
4. Type the CloudLink Center console password, renter to confirm it, and then click **Change Console Password**.

   NOTE: Step 4 is not applicable if you have already logged in to the CloudLink Center console.
5. In the **Server Name or IP Address** box, enter the DNS name or IP address that is used to connect to this server and click **Next**.

   For more information, see CloudLink Center server address.
6. Click **Generate and Download**.
7. Save the private key to an appropriate location.

   By default, the private key is saved to the download folder of your web browser.
8. Click **Next**.
9. Click **I Acknowledge**.
10. Select **Manual Unlock** or **Auto Unlock**, and then click **Next**.
11. Enter at least one CloudLink Vault passcodes and reenter to confirm it.
12. Click **Set Codes**.

Configure machine groups and volume or device encryption policy

Before deploying CloudLink Agent to machines, you may want to set up machine groups and their volume or device encryption policy. For more information, see Manage CloudLink machines by grouping.

CloudLink Center assigns machines to an existing machine group during CloudLink Agent deployment. Creating machine groups before deploying CloudLink Agent to machines enables you to assign a machine to the appropriate group during deployment. The benefit is that, after registering the machine, encryption begins automatically based on the volume encryption policy for the machine group.

If you do not specify an existing machine group during deployment, CloudLink Center assigns the machine to the Default group. By default, this group uses the manual volume or device encryption policy, which does not require encryption of any type of volume or device on machines in the group.

You can move machines to other groups after deployment. You must manually encrypt volumes based on the volume or device encryption policy for the original and new group. So that, the machine complies with the volume or device encryption policy of the new group.

For more information about creating machine groups and defining volume or device encryption policy, see the *Dell CloudLink Administration Guide*.

Set Machine Agent Upgrade policy for CloudLink Agent

If the Machine Agent Upgrade policy is set to **Auto**, connected CloudLink Agents are automatically upgraded. Any CloudLink Agent that is not connected is automatically upgraded the next time it connects to CloudLink Center.

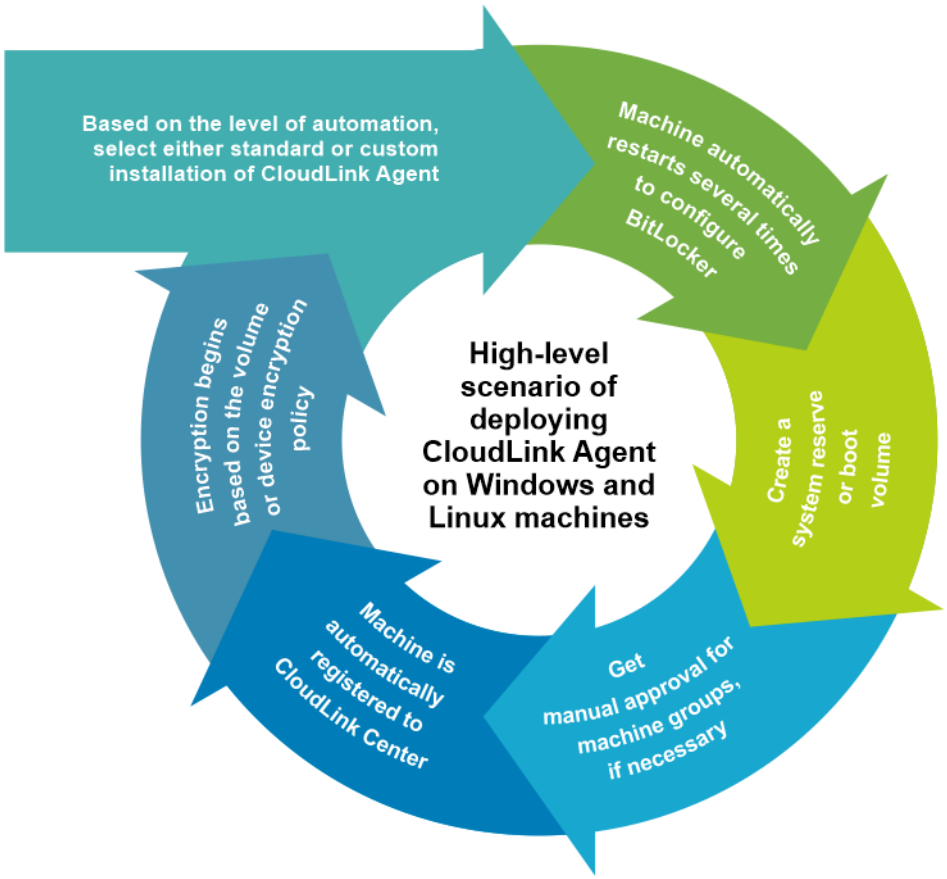
NOTE: By default, the Machine Agent Upgrade policy is set to **Auto** for the default machine group. It is recommended to set the Machine Agent Upgrade policy to **Auto**.

NOTE: In a PowerFlex Manager orchestrated environment, the PowerFlex Manager sets the Machine Agent Upgrade policy to **Manual**. Ensure that you do not change this setting.

7

Deploy, configure, and verify CloudLink Agents on Windows and Linux machines

You can deploy CloudLink Agent using a standard or custom installation. For VMs deployed in the Azure Portal, you can deploy CloudLink Agent from Azure Virtual Machine Extensions.



NOTE: For information about downloading and installing third-party signed certificates, see the "Secure CloudLink Center agents using mutual authentication" chapter in the *Dell CloudLink Administration Guide* available on the support site.

- The standard installation is an automated method that requires minimal intervention by you. It is useful for deploying CloudLink Agent to machines on an individual basis.
- The custom installation requires intervention by you, but it provides more flexibility for deployment. Unlike the standard installation, the custom installation does not automatically register the machine with CloudLink Center.

  A custom installation is useful for the following purposes:

  - Deploying CloudLink Agent to machines before deploying CloudLink Center
  - Deploying CloudLink Agent with configuration management tools

Select either the standard or custom installation that is based on the level of automation or points of manual intervention you require. At a high level, deployment includes the following processes:

- The machine might automatically restart several times to install and configure BitLocker or dm-crypt, and to create a system reserve or boot volume.
- The machine is automatically registered with CloudLink Center. This is the default setting, but a machine group could require manual approval for new machines.
- Encryption, if any, begins based on the volume or device encryption policy for the specified machine group.

For more information about the pending state and volume or device encryption policy, see the *Dell CloudLink Administration Guide*.

Manage Self-Encrypting Drives (SEDs)

See the "Manage Secure Machines" chapter in the *Dell CloudLink Administration Guide*.

Standard and Custom mode installation of CloudLink Agent on Windows and Linux

The topic describes the deployment process for each type of installation and for each OS. The following procedure enables you to determine the appropriate installation based on your requirements.

Install CloudLink Agent on Windows and Linux machines in Standard mode

NOTE:

- To install and connect an agent to CloudLink Center: See Download CloudLink Agent installer script for Standard mode on Linux CLI.
- To install agent, see Download the CloudLink Agent deployment package for Linux in Custom mode.

NOTE: To install an agent using a third-party signed certificate, see the "Secure CloudLink Center agents using mutual authentication" chapter in the *Dell CloudLink Administration Guide* available on the support site.

1. Download the CloudLink Agent installer.
2. Run the installer to complete installation and configuration.

Install CloudLink Agent on Windows machines in Custom mode

1. Download the CloudLink Agent installer package.
2. Install the package.

NOTE: If you do not want to register the machine during deployment, you can omit the CloudLink Center server address and add it manually later. Else, you can set the New Machine detection policy to get a manual approval. Registration is required for automatic encryption.

Install CloudLink Agent on Linux machines in Custom mode

1. Download the OS-specific deployment package.
2. Install the package.
3. Configure the CloudLink Center server address.

Install CloudLink Agent using Standard mode

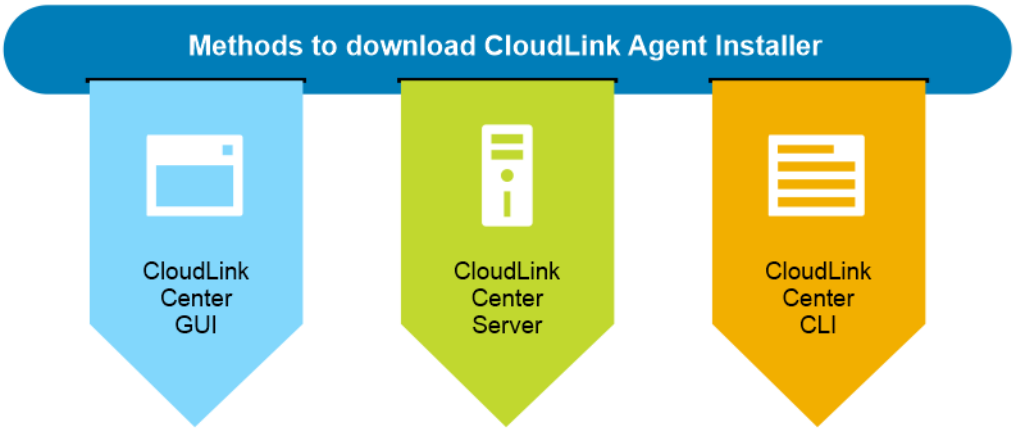
Use this procedure to install CloudLink Agent on Windows or Linux machines by using the Standard installation mode.

Steps

1. Download the installer using the CloudLink Center interface or directly from the server.
2. Run the installer from the command line to complete installation and configuration.

Download CloudLink Agent installer for Standard mode

Download the CloudLink Agent installer from CloudLink Center. For Windows, the installer is provided in the Windows installer script `clagent.bat`. For Linux, the installer is provided in the Linux installer script `clagent.sh`.



NOTE: To install and connect a third-party certificate on a Linux and Windows agent, see the "Secure CloudLink Center agents using mutual authentication" chapter in the *Dell CloudLink Administration Guide* available on the support site.

You can download the installer in one of the following methods:

- Log in to CloudLink Center and download the installer using the CloudLink Center user interface.
- Download the installer from the CloudLink Center server without logging in.
- Download the installer using a CLI.

If you are not responsible for completing the installation, provide the downloaded software to the appropriate person.

Download CloudLink Agent installer script for Standard mode by using CloudLink Center UI

Steps

1. Log in to CloudLink Center. See Access CloudLink Center.
2. From **Agents**, select **Agent Download**.
3. On the **Downloads** dialog box, select the **Windows installer script** or **Linux installer script**, and then click **Download Selected**.
4. Click **Save File**.

Results

The installer is downloaded to your download folder.

Download CloudLink Agent installer script for Standard mode directly from CloudLink Center

Steps

1. In a web browser, type the following:

   `https://clc_address/cloudlink/agent`

   Where *clc_address* is the CloudLink Center server address. The *clc_address* must be in either the FQDN, IPv4, or IPv6 format. See CloudLink Center server address.

2. Click **Save File**.

   For Linux, use the file name `clagent.sh`

Results

The installer is downloaded to your download folder.

Download CloudLink Agent installer script for Standard mode on Windows CLI

Prerequisites

To download the `clagent.bat` file by running the `Invoke-RestMethod` command, you must use the HTTPS connection. To enable a secure connection, do one of the following:

Option 1: Configure CloudLink Web TLS certificate on VM

Steps

1. On the CloudLink UI, download the CloudLink web TLS certificate to a file.
2. Add the certificate to the Trusted Root Certification Authorities certificate store either from the browser settings or using Windows Certificate Manager.
3. On Windows Server 2012 or 2012 R2 and Windows Server 2016, start a Windows PowerShell session to use TLS1.2 to communicate with the CloudLink Center web server.
4. Run the following command:

   `$TLSProtocols = [System.Net.SecurityProtocolType]'Ssl3,Tls,Tls11,Tls12'`
   `[System.Net.ServicePointManager]::SecurityProtocol = $TLSProtocols.`

   The TLS 1.2 mode is enabled only for the current PowerShell session.

   NOTE: To support TLS 1.2, use .NET Framework 4.7 and later versions.
5. Download the `Clagent.bat` installer script file by running the following command:

   ```
   Invoke-RestMethod -Uri https://clc_address/cloudlink/agent -OutFile clagent.bat
   ```

   Where, *clc_address* is the CloudLink Center server address. The *clc_address* must be in either the FQDN, IPv4, or IPv6 format. See CloudLink Center server address.

Results

The Windows installer is downloaded to the current folder.

Option 2: Ignore verification of certificate on PowerShell

1. If the CloudLink web TLS certificate is not configured on the virtual machine, run the following script at PowerShell to ignore the verification of the certificate, but only for the current PowerShell session:

```
if (-not("SkipCert" -as [type]))
{
    add-type -TypeDefinition @"
        using System;
        using System.Net;
        using System.Net.Security;
        using System.Security.Cryptography.X509Certificates;

    public static class SkipCert
    {
        public static bool RetTrue(object sender, X509Certificate certificate,
X509Chain chain, SslPolicyErrors sslPolicyErrors)
        {
            return true;
        }

        public static RemoteCertificateValidationCallback CallBack()
        {
            return new RemoteCertificateValidationCallback(SkipCert.RetTrue);
        }
    }
"@
}
```

```
$TLSProtocols = [System.Net.SecurityProtocolType]'Ssl3,Tls,Tls11,Tls12'
[System.Net.ServicePointManager]::SecurityProtocol = $TLSProtocols
[System.Net.ServicePointManager]::ServerCertificateValidationCallback =
[SkipCert]::CallBack()
```

The script downloads the `clagent.bat` file over secure channel.

2. Run the following command to download the `clagent.bat` installer script file: `Invoke-RestMethod -Uri https://<CLC_IP>/cloudlink/agent -OutFile clagent.bat`.

NOTE: You can also download the PS1 file from the following location on the support site by running the `DownloadBat.PS1 -clc <CLC IP>` command at the PowerShell interface: **https://www.dell.com/support/home/en-us/product-support/product/cloudlink-securevm/drivers**.

Download CloudLink Agent installer script for Standard mode on Linux CLI

Prerequisites

Update to the latest curl version before downloading the installation script.

NOTE: If you plan to install an agent in the third-party-certificate mode, then download the certificate file only to the `/tmp` directory. For information about downloading and installing third-party signed certificate to a Linux agent, see the "Secure CloudLink Center agents using mutual authentication" chapter in the *Dell CloudLink Administration Guide* available on the support site.

Steps

1. Use a CLI application such as wget or curl.
2. Enter one of the following commands:
   - `wget -O clagent.sh https://clc_address/cloudlink/agent`
   - `curl -o clagent.sh https://clc_address/cloudlink/agent`

   where, *clc_address* is the CloudLink Center server address. The *clc_address* must be in either the FQDN, IPv4, or IPv6 format.

Results

The Linux installer is downloaded to the current directory.

Download CloudLink Agent installer script by ignoring the verification of TLS certificate when using HTTPS

Prerequisites

None.

Option 1—Run the following command at the CLI:

Steps

1. `#wget --no-check-certificate https://cloudlink-ip-addr/cloudlink/agent -O clagent.sh`
2. `#curl --no-check-certificate https://cloudlink-ip-addr/cloudlink/agent -o clagent.sh`

   where, *cloudlink-ip-addr* is the CloudLink Center server address.

Results

The Linux installer is downloaded to the current folder.

Option 2—Run the following command at the CLI:

1. Configure the CloudLink Center Web TLS certificate on the VM by doing one of the following:
   - A) Configure the CloudLink Center Web TLS certificate on the VM by doing the following:

    - Save or import the TLS Web certificate to a file.
    - Rename the file as `cloudlink-host-name.pem`.
  - B) Enter the following command at the CLI: `#openssl s_client -servername 100.101.9.62 -connect 100.101.9.62:443 </dev/null | sed -ne '/-BEGIN CERTIFICATE-/,/-END CERTIFICATE-/p' > cloudlink-host-name.pem`
2. Export the CloudLink server IP address by running the command: `server-host-name in /etc/hosts`
3. Export the downloaded CloudLink SSL certificate to the Linux agent machine:
  - A) On SUSE machines, install the CloudLink Web TLS certificate by running the following commands:
    - `#cp cloudlink-host-name.pem /usr/share/pki/trust/anchors/`
    - `#cp cloudlink-host-name.pem /etc/pki/trust/anchors/`
    - `#update-ca-certificates`
  - B) On RHEL or CentOS machines, install the CloudLink Web TLS certificate by running the following commands:
    - `#cp cloudlink-host-name.pem /usr/share/pki/ca-trust-source/anchors/`
    - `#cp cloudlink-host-name.pem /etc/pki/ca-trust/source/anchors/`
    - `#update-ca-trust`
  - C) On Ubuntu or Debian machines, install the CloudLink Web TLS certificate by running the following commands:
    - Rename the CloudLink Web TLS certificate extension from `.pem` to `.crt`.
    - `#cp cloudlink-host-name.crt /usr/local/share/ca-certificates/`
    - `# update-ca-certificates`
4. Download the agent installation script by running the following command:
  - `#curl https://<cloudlink-host-name>/cloudlink/agent -o agent`

NOTE: Downloading of an agent script by using a Web TLS certificate works on some of Linux Distros because of different TLS library and `curl` versions. In such a case, you can use option-1 described in this procedure.

Run the CloudLink Agent installer in Standard mode

After downloading the CloudLink Agent installer from CloudLink Center, run the installer at the CLI by entering the following:

- CloudLink Center server address
- Registration code for the machine group to which you want to assign this machine (optional). See Manage CloudLink machines by grouping.

  The registration code is available from CloudLink Center on the **Agents** > **Machine Groups** page. See the *Dell CloudLink Administration Guide* .

  If you do not provide a registration code, CloudLink Center assigns the machine to the Default machine group.

Run the CloudLink Agent installer in Standard mode on Windows

About this task

On Windows VMs, CloudLink Agent automatically moves the page file to the boot drive when it is installed.

It is recommended not to move the page file to an encrypted data drive after CloudLink Agent is installed. Else, Windows cannot create administrative network shares because it uses a temporary page file.

Steps

1. In a command window, go to the folder where you downloaded the CloudLink Agent installer.

   By default, the installer is downloaded to your download folder.
2. Enter the following at the CLI:

   ```
   clagent.bat /S clc_address [/g group_code]
   ```

   Where, /S *clc_address* specifies the CloudLink Center server address. For more information, see CloudLink Center server address.

   /g *group_code* specifies the registration code for the machine group to which you want to assign this machine.

NOTE: CloudLink 7.1.3 and later versions enable you to use a third-party signed CA certificate for installing agents and connecting them to CloudLink. See the "Secure CloudLink Center agents using mutual authentication" chapter in the *Dell CloudLink Administration Guide* available on the support site.

Run the CloudLink Agent installer in Standard mode on Linux

Use this procedure to run the CloudLink Agent installer in Standard mode on Linux.

About this task

NOTE: After CloudLink Agent is installed on a Linux machine and the boot partition is encrypted; kernel upgrades and any other upgrades that involve rebuilding the kernel or initrd are not supported.

Steps

1. In a command window, go to the folder where you downloaded the CloudLink Agent installer.
   By default, the installer is downloaded to your download folder.
2. Enter the following command at the CLI:

   ```
   sudo sh clagent.sh -S clc_address [-G group_code]
   ```

   Where, -S *clc_address* specifies the CloudLink Center server address. For more information, see CloudLink Center server address.

   -G *group_code* specifies the registration code for the machine group to which you want to assign this machine.

   NOTE: CloudLink 7.1.3 and later versions enable you to use a third-party signed CA certificate for installing agents and connecting them to CloudLink. See the "Secure CloudLink Center agents using mutual authentication" chapter in the *Dell CloudLink Administration Guide* available on the support site.

Scenario-based limitations for installing the CloudLink Agent installer in Standard mode by using IPv6 protocol

Some of the limitations you have when installing CloudLink Agent, in standard mode by using IPv6 protocol, are listed here:

Install CloudLink Agent by using an IPv6 link-local address without scope

Windows client

- Installation—Not allowed
- Example—`clagent.bat /S fe80::0`
- Sample error message: IPv6 link local address must include a scope ID.

Linux client

- Installation—Not allowed
- Example—`sh clagent.sh -S fe80::0`
- Sample error message: IPv6 Link local address `<IPv6 address>` requires scope.

Install CloudLink Agent by using an IPv6 link-local address with scope indicated by the network interface ID

Windows client

- Installation—Not allowed
- Example—`clagent.bat /S fe80::0%12`
- Sample error message: Not applicable

Linux client

- Installation—Not allowed
- Example—`sh clagent.sh -S fe80::0%2`
- Sample error message: IPv6 link local address scope can be interface name only.

Install CloudLink Agent by using an IPv6 link-local address with scope indicated by the network interface name

Windows client

- Installation—Not allowed
- Example—`clagent.bat /S fe80::0%Ethernet`
- Sample error message: IPv6 link local address scope can be interface number only.

Linux client

- Installation—Allowed
- Example—`sh clagent.sh -S fe80::0%eth0`
- Sample error message: Not applicable

Install CloudLink Agent using IPv6 link-local address on Debian 10 operating system

The following procedure explains how to install CloudLink Agent by using an IPv6 link-local address on Debian 10 operating system.

About this task

NOTE: You can use this procedure to install CloudLink Agent using IPv6 link-local address on Debian 10 operating system using Standard and Custom mode.

To download and run the CloudLink Agent on Debian 10 operating system, type the following commands:

Steps

1. To list all the IP addresses associated a network interface, type the following command:
   `ip a ens192`
   where, ens 192 is the name of the network interface.
2. `curl --interface ens192 [IPv6-link-local address ]/cloudlink/securevm > securevm`
3. Specify the scope of the IPv6 link-local address. Interface name is the scope of the link-local address.
   For example: `sh securevm -S [IPv6 link local address]%ens192`

Download and Install CloudLink Agent on Windows and Linux using Custom mode

This topic describes the procedure to download and install CloudLink Agent on Windows and Linux using Custom mode of installation.

Download and run CloudLink Agent using Custom mode on Windows

Use this procedure to download and run CloudLink Agent using the Custom mode on Windows.

Steps

1. Download the CloudLink Agent installer package. See, Download CloudLink Agent installer package for Windows in Custom mode.
2. Run the installer from the CLI to complete the installation and configuration.

Download CloudLink Agent installer package for Windows in Custom mode

Use this procedure to download the CloudLink Agent installer package for Windows in Custom mode.

Steps

1. Log in to CloudLink Center.
2. From **Agents**, select **Agent Download**.
3. From the **Downloads** page, select the **64-bit Windows Installer package** (`securevm-windows-x64.msi`), and then click **Download Selected**.
4. Click **Save File**.

Results

The installer package is downloaded to your Downloads directory.

Run the CloudLink Agent installer package in Custom mode on Windows

Use this procedure to run the CloudLink Agent installer package on Windows using Custom mode.

About this task

After downloading the CloudLink Agent installer package (the `securevm-windows-x64.msi` file) from CloudLink Center, you can run it from the CLI or by using Windows Installer tools.

Steps

1. Go to the folder where the CloudLink Agent installer package is located.
2. From the command line, type the following:

```
msiexec /i securevm-windows-x64.msi [CLOUDLINKCENTER=clc_address]
[GROUPCODE=group_code]
```

NOTE: CloudLink 7.1.3 and later versions enable you to use a third-party signed CA certificate for installing agents and connecting them to CloudLink. See the "Secure CloudLink Center agents using third-party signed certificates" chapter in the *Dell CloudLink Administration Guide* available on the support site.

- CLOUDLINKCENTER=*clc_address* specifies the CloudLink Center server address. For more information, see CloudLink Center server address. If you do not specify the CloudLink Center server address, the machine is not registered with

CloudLink Center. No automatic encryption occurs after deployment. For information about specifying this address following deployment, see Add the CloudLink Center server address or group registration code after deployment.

- GROUPCODE=*group_code* specifies the registration code for the machine group to which you want to assign the machine. If you do not specify a registration code, CloudLink Center assigns the machine to the Default machine group. For more information, see Manage CloudLink machines by grouping.

  You obtain the machine group registration code from CloudLink Center. The code is displayed on **Agents** > **Machine Groups**. For more information, see the *Dell CloudLink Administration Guide*.

3. When the CloudLink Agent Setup Wizard is displayed, click **Install**.
4. Click **Finish**.
5. Wait for the installation to complete.

Results

The machine automatically restarts one or more times.

Add the CloudLink Center server address or group registration code after deployment

About this task

As a deployment option, you might want to omit the CloudLink Center server address when running the `MSI` file from the CLI. Similarly, you might not have access to the group registration code required to assign a machine to a machine group other than **Default**. See, Run the CloudLink Agent installer package in Custom mode on Windows.

Steps

Run the following command to add the CloudLink Center server address or group registration code:

```
svm /S clc_address [/g group_code]
```

Where, /S *clc_address* specifies the CloudLink Center server address. For more information, see CloudLink Center server address.

/g *group_code* specifies the registration code for the machine group to which you want to assign this machine.

Verify successful deployment of CloudLink Agent on Windows machines in Custom mode

Verify successful deployment of CloudLink Agent on Windows machines by using one of the following.

- Log in to CloudLink Center, and view the machine status. For information about managing machines, including viewing their status, see the *Dell CloudLink Administration Guide*.
- In the Windows taskbar, ensure that the  icon is displayed. The tooltip displays a message indicating that the machine is connected to CloudLink Center.

Download, install, and configure CloudLink Agent using Custom mode on Linux machines

Use this procedure to download, install, and configure CloudLink Agent using custom mode on Linux machines.

Steps

1. Download the CloudLink Agent deployment package.
2. Install the CloudLink Agent deployment package.
3. Configure CloudLink Agent.

Download the CloudLink Agent deployment package for Linux in Custom mode

Use this procedure to download CloudLink Agent deployment package for Linux machines in Custom mode.

Prerequisites

Update to the latest curl version before downloading the installation script.

About this task

CloudLink Agent deployment packages are available as `RPM`, `TGZ`, or `DEB` files which you download from CloudLink Center.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Agent Download**.
3. On the **Downloads** page, select the 64-bit Linux package that you want to use. The package URL is displayed below the selected package.
   For example,

   ```
   securevm.centos.x86_64.rpm
   ```

4. In the Linux machine, open a command-line client and enter one of the following commands:

   ```
   wget https://clc_address/cloudlink/agent/url_of_package
   ```

   ```
   curl -O https://clc_address/cloudlink/agent/url_of_package
   ```

   where, *clc_address* is the CloudLink Center server address, and *url_of_package* is the package URL. The *clc_address* must be in either the FQDN, IPv4, or IPv6 format.

   For example,

   ```
   wget https://192.168.112.157/cloudlink/agent/securevm.centos.x86_64.rpm
   ```

Results

The deployment package is downloaded to the Linux machine to be encrypted.

Install the CloudLink Agent deployment package for Linux in Custom mode

About this task

After downloading the deployment package for your operating system from CloudLink Center, do the following.

Steps

Install the package using the package manager for your platform.

NOTE: After CloudLink Agent is installed on a Linux machine and the boot partition is encrypted, kernel upgrades and any upgrades that involve rebuilding the kernel or initrd are not supported.

Configure CloudLink Agent on Linux machines

Use this procedure to configure CloudLink Agent on Linux machines using Custom mode.

About this task

The deployment package installation installs the CloudLink Agent, which provides the `svm` subcommand for configuring CloudLink Agent. During configuration, the machine is registered with CloudLink Center.

NOTE: CloudLink 7.1.3 and later versions enable you to use a third-party signed CA certificate for installing agents and connecting them to CloudLink. See the "Secure CloudLink Center agents using mutual authentication" chapter in the *Dell CloudLink Administration Guide* available on the support site.

Steps

1. Enter the following command to configure CloudLink Agent.

   ```
   svm [-v ] [-G group_code]-S clc_address
   ```

   Where,

   - -v uses verbose mode.
   - -G *group_code* specifies the registration code for the machine group that you want to assign this machine to.
   - -S *clc_address* specifies the CloudLink Center server address. The *clc_address* must be in either the FQDN, IPv4, or IPv6 format. For more information, see CloudLink Center server address.

2. Restart the machine.

Verify successful deployment of CloudLink Agent on Linux machines in Custom mode

About this task:

Verify CloudLink Agent deployment from the machine CLI and encryption status of volumes or devices by entering the following command:

```
svm status
```

Refresh data after deployment of CloudLink Agent on Linux machines in Custom mode

About this task

For Linux machines, if the networking configuration is changed on the client after CloudLink Agent deployment, refresh the CloudLink Agent service.

Steps

Refresh the CloudLink Agent service from the machine CLI by entering the following command:

```
svm refresh
```

CloudLink Agent for Microsoft SQL Server

If data drives encrypted by CloudLink Agent on a Windows machine running Microsoft SQL Server gets locked, SQL Server cannot load database files on those data drives.

Using PowerShell commands, the start of SQL Server services can be delayed until the data drive is available. After the CloudLink Agent unlocks the encrypted data drive, it can manually start SQL Server services that are dependent on CloudLink.

After CloudLink Agent is uninstalled, all dependent services revert to their previous state.

BitLocker auto-unlock feature

As an alternative to delaying the start of SQL Server services, you can use the auto-unlock feature to unlock data drives during the Windows boot process. To automatically enable auto-unlock, ensure that the boot drive is encrypted BEFORE you encrypt the data drive.

Dependencies of Microsoft SQL Server on CloudLink Agent

PowerShell commands:

- Make the SQL Server service dependent or independent of CloudLink Agent.
- Encrypt or decrypt the SQL Server data drive.

The Following are PowerShell commands:

- To view the volume encryption status, run the `svm status` command.
- To encrypt a SQL server data drive, run the `svm encrypt [volume_letter]` command.
- To decrypt a SQL server data drive, run the `svm decrypt [volume_letter]` command.
- To set the SQL Server service to depend on CloudLink, run the `svm setdeps <Microsoft SQL Server service name>` command.

  For example, `svm setdeps MSSQLServer`.
- To view the list of services that depend on CloudLink, run the `svm showdeps` command.
- To remove the SQL Server service from the CloudLink dependency list, run the `svm cleardeps <Microsoft SQL Server service name>` command.

  For example, `svm cleardeps MSSQLServer`.

Deploy CloudLink Agent Azure extension using the Azure portal

Use the following procedure to install CloudLink Agent on a VM in the Azure Portal using an Azure VM extension.

Steps

1. In the Azure Portal, click **Create a resource**.
2. In the **New** blade, select your VM.
3. Follow the prompts to configure your VM until you get to the **Guest config** page.
4. From the **Guest config** page, click **Select an extension to install**.
5. From the extension list, select **CloudLink SecureVM Agent**.
6. Click **Create**.
7. In the **CloudLink Center Address** box, enter the CloudLink Center address in either the FQDN or IPv4 format.

   This field is mandatory.
8. In the **Machine Group Registration Code** box, enter the machine group registration code to register the VM with that particular machine group.

   This field is optional.

   NOTE: If the machine group registration code is not specified, the VM registers with the default machine group in CloudLink Center.
9. Click **OK**.
10. Continue to create your VM.
11. Wait for the VM to appear in the CloudLink Center **Machines** pane.

Results

After CloudLink Agent is installed, you can:

- To view the CloudLink Agent extension installation status, extension version, and other details, check the Extensions settings of the VM in the Azure Portal.
- Log in to the VM to ensure successful installation.

Deploy CloudLink Agent Azure extension to a Windows machine using PowerShell

Use this procedure to install CloudLink Agent on a Windows machine that was deployed using the Azure Resource Manager.

Prerequisites

- Azure PowerShell must be installed on the system you are using to manage your Azure resources.
- The AzureRM module for PowerShell must be installed and loaded.

Steps

1. Start PowerShell and log in to your Azure account with the following commands.
   a. Enter the following command to log in to your Azure account:

   ```
   Login-AzureRmAccount
   ```

   b. Enter your Azure account username and password when you are prompted.
   c. Enter the following command to add your Azure account:

   ```
   Add-AzureAccount
   ```

   d. Enter your Azure account username and password when you are prompted.
2. Enter the following commands to specify your Windows machine name, resource group, and location.

   ```
   $vmname = “YourVMName” $resourceGroup = “YourVMResourceGroup” $location =
   “YourVMLocation”
   ```

   Example:

   ```
   $vmname = "Test-win-2012r2" $resourceGroup = "Test-ResourceGroup" $location = "East
   US"
   ```

3. Enter the following commands to specify the CloudLink Agent extension name, namespace, and version.

   ```
   $extname = “CloudLinkSecureVMWindowsAgent” $nameSpace = “CloudLinkEMC.SecureVM”
   $extversion = (Get-AzureVMAvaialableExtension -ExtensionName $extname).Version
   ```

4. Enter the following command to specify the CloudLink Center IP address and machine group registration code.

   ```
   $SettingsString = ‘{“CloudLinkCenter” :”X.X.X.X”, “GroupRegistrationCode” : “XXXX-
   XXXX”}’
   ```

   NOTE: The *GroupRegistrationCode* parameter is optional. If it is not specified, the VM is registered to the default machine group in CloudLink Center.

   Example:

   ```
   $SettingsString = ‘{“CloudLinkCenter” :”192.168.1.1”, “GroupRegistrationCode” : “53ad-
   cc66”}’
   ```

5. Enter the following command to add the CloudLink Agent extension to the Windows machine:

```
Set-AzureRmVMExtension -ResourceGroupName $resourceGroup -Location $location -VMName
$vmname -Name $extname -Publisher $nameSpace -Type $extname -TypeHandlerVersion
$extversion -SettingString $SettingsString
```

6. Wait for the Windows machine to appear in the CloudLink Center **Machines** pane.

Results

After CloudLink Agent is installed:

- To see the CloudLink Agent extension installation status, the extension version, and other details, click the Extension settings of the Windows machine in the Azure Portal .
- Log in to the Windows machine to confirm successful installation.

Deploy CloudLink Agent Azure extension to a Linux machine using PowerShell

Use this procedure to install CloudLink Agent on a Linux machine that was deployed using the Azure Resource Manager.

Prerequisites

- Azure PowerShell must be installed on the system you are using to manage your Azure resources.
- The AzureRM module for PowerShell must be installed and loaded.

Steps

1. Start PowerShell and log in to your Azure account with the following commands:
   a. Enter the following command to log in to your Azure account:

   ```
   Login-AzureRmAccount
   ```

   b. Enter your Azure account username and password when you are prompted.
   c. Enter the following command to add your Azure account:

   ```
   Add-AzureAccount
   ```

   d. Enter your Azure account username and password when you are prompted.
2. Enter the following commands to specify your Linux machine name, resource group, and location.

   ```
   $vmname = "YourVMName" $resourceGroup = "YourVMResourceGroup" $location =
   "YourVMLocation"
   ```

   Example:

   ```
   $vmname = "Test-rhel-7" $resourceGroup = "Test-ResourceGroup" $location = "East US"
   ```

3. Enter the following commands to specify the CloudLink Agent extension name, namespace, and version.

   ```
   $extname = "CloudLinkSecureVMLinuxAgent" $nameSpace = "CloudLinkEMC.SecureVM"
   $extversion = (Get-AzureVMAvaialableExtension -ExtensionName $extname).Version
   ```

4. Enter the following command to specify the CloudLink Center IP address and machine group registration code.

   ```
   $SettingsString = '{"CloudLinkCenter" :"X.X.X.X", "GroupRegistrationCode" : "XXXX-
   XXXX"}'
   ```

NOTE: The *GroupRegistrationCode* parameter is optional. If it is not specified, the VM is registered to the default machine group in CloudLink Center.

Example:

```
$SettingsString = '{"CloudLinkCenter" :"192.168.1.1", "GroupRegistrationCode" : "53ad-cc66"}'
```

5. Enter the following command to add the CloudLink Agent extension to the Linux machine:

```
Set-AzureRmVMExtension -ResourceGroupName $resourceGroup -Location $location -VMName $vmname -Name $extname -Publisher $nameSpace -Type $extname -TypeHandlerVersion $extversion -SettingString $SettingsString
```

6. Wait for the Linux machine to appear in the CloudLink Center **Machines** pane.

Results

After CloudLink Agent is installed:

- To see the CloudLink Agent extension installation status, the extension version, and other details, click the Extension settings of the Linux machine in the Azure Portal .
- Log in to the Linux machine to confirm successful installation.

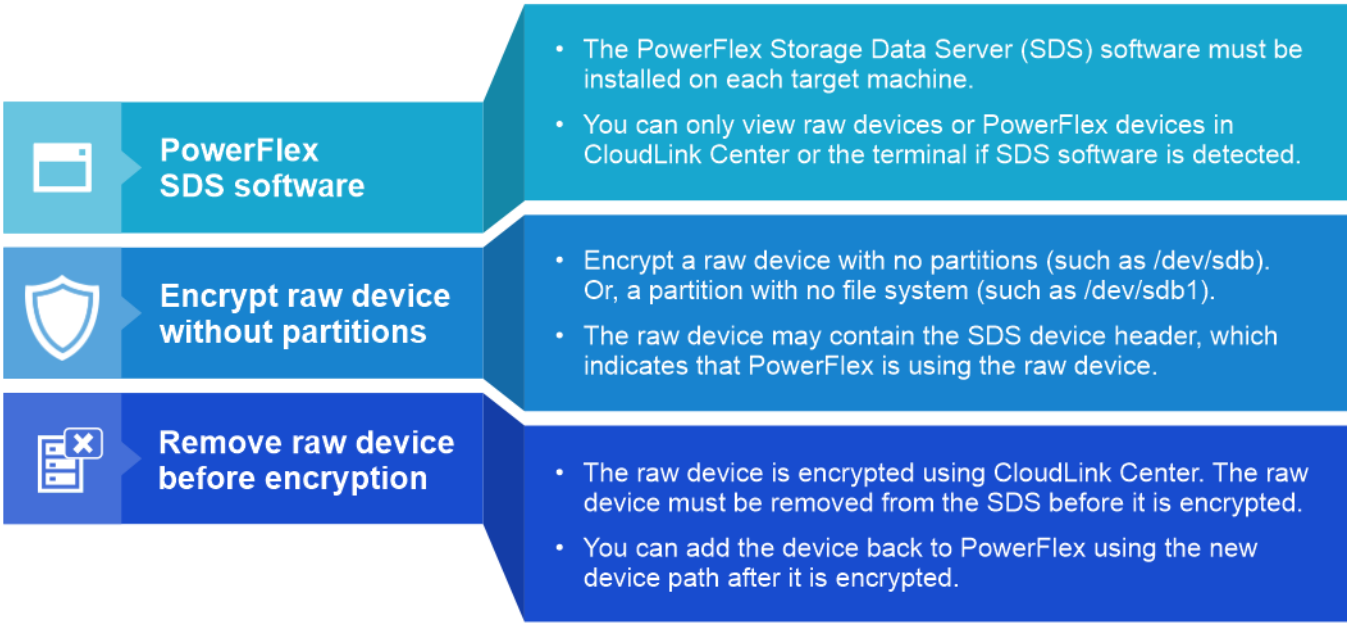
8

Using PowerFlex devices

This chapter presents the following topics:

Requirements to encrypt PowerFlex devices

This topic provides information about the requirements to encrypt PowerFlex devices.



NOTE: The device mapping name changes after the device is encrypted.

Encrypt a new PowerFlex device

The following procedure explains how to encrypt a new PowerFlex device.

About this task

The Linux virtual machine must have CloudLink Agent installed, and it must be registered with CloudLink Center.

Steps

1. In CloudLink Center, click **Agents** > **Machines**.
2. Select the SVM node that you want to add to PowerFlex.
3. Log in to the PowerFlex user interface and go to **Block** > **Storage System**
   The **Storage System | Protection Domains** list page is displayed.
4. Select the storage device from the list and click **Add Storage System**.
   The **Add Storage Device** window is displayed.
5. Enter the path where the device is located and the name of the device.
6. Select the storage pool to which you want to add the device and click **Add Devices**.
7. Encrypt the raw device.
8. Copy the encrypted device name.

For example, `/dev/mapper/svm_sdb`.

9. Log in to the PowerFlex user interface and go to **Block** > **Storage System**
   The **Storage System | Protection Domains** list page is displayed.
10. Select the storage device from the list and click **Add Storage System**.
    The **Add Storage Device** window is displayed.
11. Enter the path where the device is located and the name of the device.
12. Select the storage pool to which you want to add the device and click **Add Devices**.
13. Type or paste the device name and click **OK**.

    PowerFlex begins a rebalance operation and adds the device.

    When the device is added to PowerFlex, CloudLink Center detects the PowerFlex storage node device header, and the device type is changed to storage node. You can also use the PowerFlex CLI tools or REST APIs to add the device to the pool.

Encrypt an existing PowerFlex device using PowerFlex Manager

The following steps describe the procedure to encrypt an existing device using PowerFlex Manager.

About this task

Perform the following steps in the PowerFlex Manager user interface to encrypt an existing device:

Steps

1. Using the PowerFlex UI, remove the device from PowerFlex.
2. Wait until the device is removed from PowerFlex.

   The device type is changed to an unencrypted raw device in CloudLink Center.

3. In CloudLink Center, click **Agents** > **Machines**.
4. Select the device that you want to encrypt.
5. In the **Actions** menu, select **Encrypt**.

   CloudLink Center encrypts the device.

6. Copy the encrypted device name.

   For example, `/dev/mapper/svm_sdb`

7. Log in to the PowerFlex Manager user interface and go to **Lifecycle** > **Storage Nodes** .

   You can also go to **Block** > **Storage System** and perform the following steps:
   a. Select the storage device from the list and click **Add Device**. The **Add Storage Device** window is displayed.
   b. Enter the path where the device is located and the name of the device.
   c. Select the storage pool to which you want to add the device and click **Add Devices**.
   d. Type or paste the device name, and then click **OK**. PowerFlex begins a rebalancing operation and adds the device.

   The **Storage Nodes** list page is displayed.
8. Select the required storage node from the list and click **More Actions** > **Add Devices**.
   The **Add Device to Storage Node <storage node name>** window is displayed.
9. Enter the path and name of the device and select the device group from the **Device Group** dropdown.
10. Click **Add devices**. PowerFlex adds the device.

Manage PowerFlex devices from the command line

Run the following commands to manage PowerFlex devices from the CLI.

- Enter the following command to log in to the MDM:

  ```
  scli -login -username=admin
  ```

- Enter the following command to encrypt a PowerFlex device:

  ```
  svm encrypt /dev/sd X
  ```

  where, *X* is the drive letter.

- Enter the following command to erase a PowerFlex device:

  ```
  svm erase /dev/sd X
  ```

  where, X is the device letter.

- Enter the following command to add a device to the SDS:

  ```
  scli --add_sds_device --storage_pool_name <EX-POOL-NAME>
   --sds_name <EX-SDS-NAME> --device_path /dev/mapper/svm_sd X --device_name svm_sd X
  ```

  where, *EX-POOL-NAME* and *EX-SDS-NAME* are your storage pools.

  SDS names and X is the device letter.

  `EX-SDS-NAME` can also be an IP address.

9

Uninstall CloudLink Agent

This chapter describes the procedures to uninstall CloudLink Agent on Windows, and Linux machines.

If you are uninstalling the storage node device package from the PowerFlex storage node, it is recommended to uninstall the CloudLink Agent first. For more information, see the "Erase a PowerFlex storage node device" section in the *Dell CloudLink Administration Guide*.

Uninstall CloudLink Agent on Windows

Use this procedure to uninstall CloudLink Agent on Windows.

Steps

1. Decrypt any encrypted volumes on the machine.
2. Use the `Msiexec.exe` tool for uninstalling applications.

Uninstall CloudLink Agent on Linux

Use this procedure to uninstall CloudLink Agent on Linux.

Steps

1. Decrypt any encrypted volumes on the machine.
2. Enter the following command:
   ```
   svm uninstall
   ```
3. If you installed CloudLink Agent using Azure VM Extensions through PowerShell, remove CloudLink Agent using PowerShell.

10

Troubleshooting

CloudLink installation problems and workarounds

This topic describes the CloudLink installation problems and possible workarounds.

CloudLink Azure extension is uninstalled, but CloudLink Agent software is not getting uninstalled

Description	If you uninstall the CloudLink Azure Extension for a machine that has encrypted volumes, the CloudLink Azure Extension is uninstalled. But, CloudLink Agent software is not getting uninstalled.
Workaround	Decrypt the encrypted volumes before uninstalling the CloudLink Azure extension.

DNS configuration is lost after you restore CloudLink Center from a backup file

Description	If you restore CloudLink Center from a backup file, then the DNS configuration may be lost.
Workaround	Add the DNS configuration manually, if it is missing in CloudLink Center.

CloudLink Center fails to obtain an IP address from DHCP

Description	CloudLink Center may fail to obtain an IP address from DHCP.
Workaround	This issue does not occur frequently. If it occurs, select **Network Reconfigure** from the CloudLink Center console to acquire a new IP address.

If a CloudLink Center instance with dual NICs is restarted, the default gateway moves to the secondary NIC

Description	If a CloudLink Center instance with dual NICs that use DHCP for IPv4, is restarted, then the default gateway may move to the secondary NIC.
Workaround	Do not use DHCP for IPv4 with both NICs.

CloudLink Agent installation problems and workarounds

This topic describes the known CloudLink Agent installation problems and limitations, and possible workarounds.

VM experiences a system failure during startup

Description	If the Active Directory domain services database, log files, and SYSVOL files are on an encrypted data volume, the machine experiences a system failure during startup.
Workaround	Ensure that the Active Directory domain services database, log files, and SYSVOL files are on the boot volume.

Windows machine fails to reboot

Description	Windows machine fails to reboot when using EFI bootloader, and the machine is connected to CloudLink Center using secondary NIC in a two-adapter configuration.
Workaround	Use the primary NIC to connect to CloudLink Center.

CloudLink agent is installed but fails to register on the CloudLink Center

Description	If you misprinted Machine Group Registration Code parameter during CloudLink Agent Azure Extension installation on a Linux VM, the CloudLink agent is installed, but fails to register on the CloudLink Center.
Workaround	Uninstall CloudLink Agent Azure Extension, and then install the CloudLink Agent Azure Extension using correct set of parameters.

Encryption for Containers problems and workarounds

This topic describes the known Encryption for Containers problems and limitations, and possible workarounds.

Invalid CloudLink Center address list or inaccessible addresses are displayed

Description

```
kubectl -n sdp logs demo-cloudlink-sdp-nfs-encrypted-2lg4z
```

proxy logs displays invalid CloudLink Center address list or inaccessible addresses.

Workaround Verify that the secret for the corresponding cluster that is created in the CloudLink Center is applied to the Kubernetes cluster.

Container pods stops functioning

Description The container pods stops functioning, and displays `/var/lib/kubelet/plugins/csi-sdp-vxflexos/csi-vxflexos.sock` is already in use error message.

Workaround	Delete the socket file on each of the worker node at `/var/lib/kubelet/plugins/csi-sdp-vxflexos/csi-vxflexos.sock`.

Encryption for containers logs displays driver is not authenticated error message for PowerFlex

Description	The `kubectl logs -n vxflexos demo-cloudlink-sdp-vxflexos-encrypted-ctl-0` driver logs displays the "driver is not authenticated" error message for PowerFlex.
Workaround	Verify the following for the PowerFlex system: • Username • Password • Gateway IP address

Encryption for containers driver logs displays driver error message

Description	The `kubectl logs -n Isilon demo-cloudlink-sdp-isilon-encrypted-ctl-0` driver logs displays the "Create volume failed, access denied. Create directory as requested" error message. This occurs when the user who created the base path is different from the user who configured the driver.
Workaround	Ensure that the user who deploys the driver has sufficient rights on the base path (isiPath) to perform the required operations.

Encryption for containers logs displays driver is not authenticated error message for PowerScale

Description	The `kubectl logs -n Isilon demo-cloudlink-sdp-isilon-encrypted-ctl-0` driver logs displays the "driver is not authenticated" error message.
Workaround	Verify the username and the password of the secret for the corresponding cluster.

Pod status in kubectl is ImagePullBackOff in place of running

Description	Pod status in kubectl is ImagePullBackOff in place of running. This occurs if the docker registry is unreachable or if the image is not available.
Workaround	Verify the following: • Docker registry is reachable. • Image is available.

Key Management Interoperability Protocol (KMIP) server issues and workarounds

This topic describes the known Key Management Interoperability Protocol (KMIP) server issues, limitations, and possible workarounds.

Several nodes associated with CloudLink are not functioning

Description	A few nodes that are associated with CloudLink have stopped functioning but at least one node is still functioning.
Impact	No impact. The ESXi vSAN environment remains functional even if one node is functioning.
Solution	Contact Dell Technologies support. If the non-functional nodes cannot be returned to normalcy, do the following: ● Redeploy the non-functional nodes from the OVA. ● Reconfigure the node as done in the initial build process. ● Add the redeployed node to the existing cluster.

Cluster is not functioning and unresponsive

Description	The cluster is not functioning and unresponsive to vSAN. However, the cluster status can be restored.
Impact	ESXi vSAN continues to function. However, when an ESXi host is restarted, it fails to retrieve its Key Encryption Key (KEK) and mount the disks. ESXi vSAN redundancy prevents data loss, but affects capacity and performance.
Workaround	Restore the status of the cluster so the ESXi hosts can retrieve their KEKs.

Cluster is not functioning and cannot be restored

Description	Cluster is not functioning and its status cannot be restored.
Impact	ESXi vSAN continues to function. However, when an ESXi host is restarted, it fails to retrieve its Key Encryption Key (KEK) and mount the disks. ESXi vSAN redundancy prevents data loss, but affects capacity and performance. Data loss may occur if the outage was prolonged and multiple ESXi hosts failed to respond before CloudLink Center was restored.
Solution	If the cluster status cannot be restored, restore the cluster from the backup file. Also, do the following: ● Rebuild a single Cloudlink node VM by using the OVA. ● Restore the cluster from the backup file. This restores the contents of the key database and cluster starts functioning. Redeploy the remaining nodes from the OVA template and add the nodes to the restored cluster.

Cluster is not functioning and cannot be restored from a backup file

Description	Cluster is not functioning and cannot be restored from a backup file.
Impact	ESXi vSAN continues to function. However, when an ESXi host is restarted, it fails to retrieve its Key Encryption Key (KEK) which affects the capacity and performance. Because the keys are unrecoverable, recovery actions must occur before an ESXi host is restarted. Data loss may occur as multiple ESXi hosts are restarted, or errors are detected before KMIP services are restored.

Solution

Do the following:

1. Redeploy a new CloudLink cluster by using the OVA template.
2. Create KMIP partitions and KMIP client IDs for each vCenter client connection.
3. Configure vCenter with a new Key Management Server (KMS) provider, directing to the new CloudLink cluster.
4. Set the new CloudLink cluster as the default KMS provider in vCenter.

Because the existing keys are still available in the ESXi host memory and the data is accessible, perform a shallow rekey operation on the VSAN cluster using the new KMS provider without losing the existing data. Old keys are no longer required after the shallow rekey operation is completed.

Migrate KMIP clients because cluster is configured to incorrect KMIP client ID or cluster

Description

KMIP clients should be migrated from an existing CloudLink cluster because a cluster is configured to incorrect KMIP client ID or cluster, workloads, or change of plan.

Impact

Migrate the client vSAN to a different Cloudlink client ID, or cluster without interrupting the CloudLink service.

Solution

Do the following:

1. Configure a new KMS definition on the required vCenter server using the target CloudLink cluster and client information.
2. Set the default KMS provider to the newly defined KMS connection.
3. Perform a shallow rekey operation of the cluster using the new KMS provider.
4. Remove the old KMS provider after all the clients of the old KMS are rekeyed.

   Contact VMware support to ensure that old KMS is no longer in use.

11

Appendix—Reference topics

CloudLink Center server address

You use the CloudLink Center server address frequently. For example, you provide the address in the URL used to access the CloudLink Center user interface and in commands that are used to download installation files.

You can specify the CloudLink Center server address in one of the following formats:

- **IP address default**

  Enterprise and PowerFlex—It is recommended that you use a static IPv4 address. For IPv6, CloudLink only supports dynamic address assignment.

  NOTE: If you use IPv6, an IPv4 address must be assigned to the CloudLink Center network interface using either static or DHCP addressing. CloudLink only supports dual-stack devices.

  Microsoft Azure and Azure Stack—You can use an IPv4 address. It is recommended using the DNS name because IP address of CloudLink Center can change after power cycling.

  NOTE: The server address can be either the public external address or the private internal address. The choice depends on the location of the CloudLink Agents connecting to CloudLink Center. The external DNS name is the best choice usually.

- **Hostname**

  If the Domain Name System (DNS) has an entry for CloudLink Center, it is recommended that you specify the CloudLink Center server address as a hostname in fully qualified domain name (FQDN) format, such as clc.example.com. For more information, see the *Dell CloudLink Administration Guide*.

Requirements for CloudLink Center server addresses in clusters

In a CloudLink Center cluster, servers and CloudLink Agents use the CloudLink Center server address for communication. You define this address, as the *Server Name/Address*, when deploying a new server. You can use either the IP address or hostname (FQDN format).

NOTE: Only use the hostname in FQDN format or the external DNS name for Azure and Azure Stack machines with dynamic IP addresses.

Ensure that you specify the server address using the preferred format for each CloudLink Center server before creating the cluster. You can use FQDNs and IP addresses in a cluster, but you cannot change the server address format after creating a cluster.

Prestartup authorization of CloudLink Center machines

Prestartup authorization of CloudLink Center machines applies to only Enterprise and Microsoft Azure and Azure Stack.

Prestartup authorization enables a machine to start automatically when the machine has been previously registered with CloudLink Center and can connect to it.

NOTE: If the boot volume of a machine is not encrypted, but one or more data volumes are encrypted, the machine is allowed to start. After the machine starts, CloudLink Center determines whether encryption keys for encrypted data

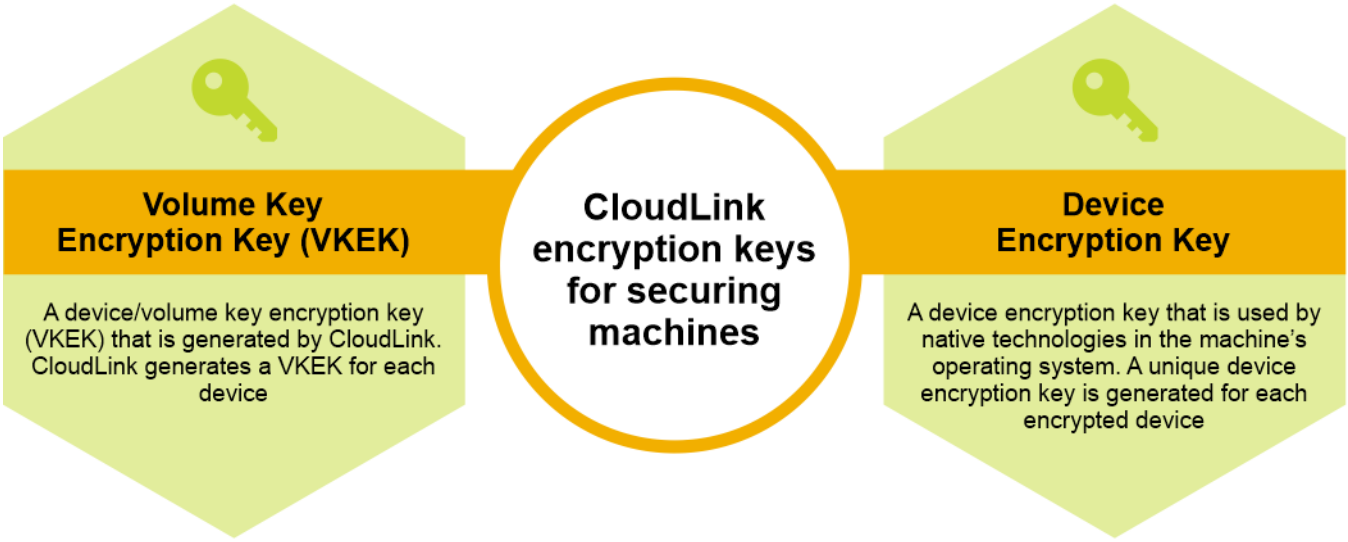
volumes can be released automatically based on key release policies. If key release policies are not met for the data volume, CloudLink Center puts the machine in the pending state.

If a machine does not pass prestartup authorization, CloudLink Center puts the machine in the pending state and you must explicitly accept the machine before startup is allowed to continue.

For information about approved networks, removing or blocking a machine, and the pending state, see the *Dell CloudLink Administration Guide*.

Secure CloudLink machines by using encryption keys

CloudLink uses the following types of encryption keys to secure machines:



These keys are stored in CloudLink Center or another keystore. For more information, see Encryption key location and protection options in CloudLink.

For a machine, key release policies determine the volume or device encryption keys to secure the boot or data volumes, or encrypted devices. For more information, see CloudLink Key Release policies. The VKEK pair protects the volume encryption keys:

- When CloudLink Center receives a request from CloudLink Agent to encrypt a volume or a device on its machine, CloudLink Center generates a new VKEK in the current keystore and uses it to encrypt the volume or device encryption key.
- When a volume requires decryption, CloudLink Center decrypts the volume encryption key using the VKEK and sends it to CloudLink Agent.

There is a difference between the types of encryption keys that are used to secure machines because native technologies in the operating system of the machines create and manages volume or device encryption keys.

CloudLink documentation does not discuss keys in detail. Unless specified otherwise, the terms *encryption keys* and *keys* in this documentation refer to VKEKs.

CloudLink Key Release policies

This topic provides information about the key release policies available in CloudLink.

Before CloudLink Center automatically releases keys, a machine must:

- Fulfill the requirements of key release policies
- Use an IP address that belongs to an approved network
- Belong to an approved location
- Not been removed previously

Key release policies may be required to enable:

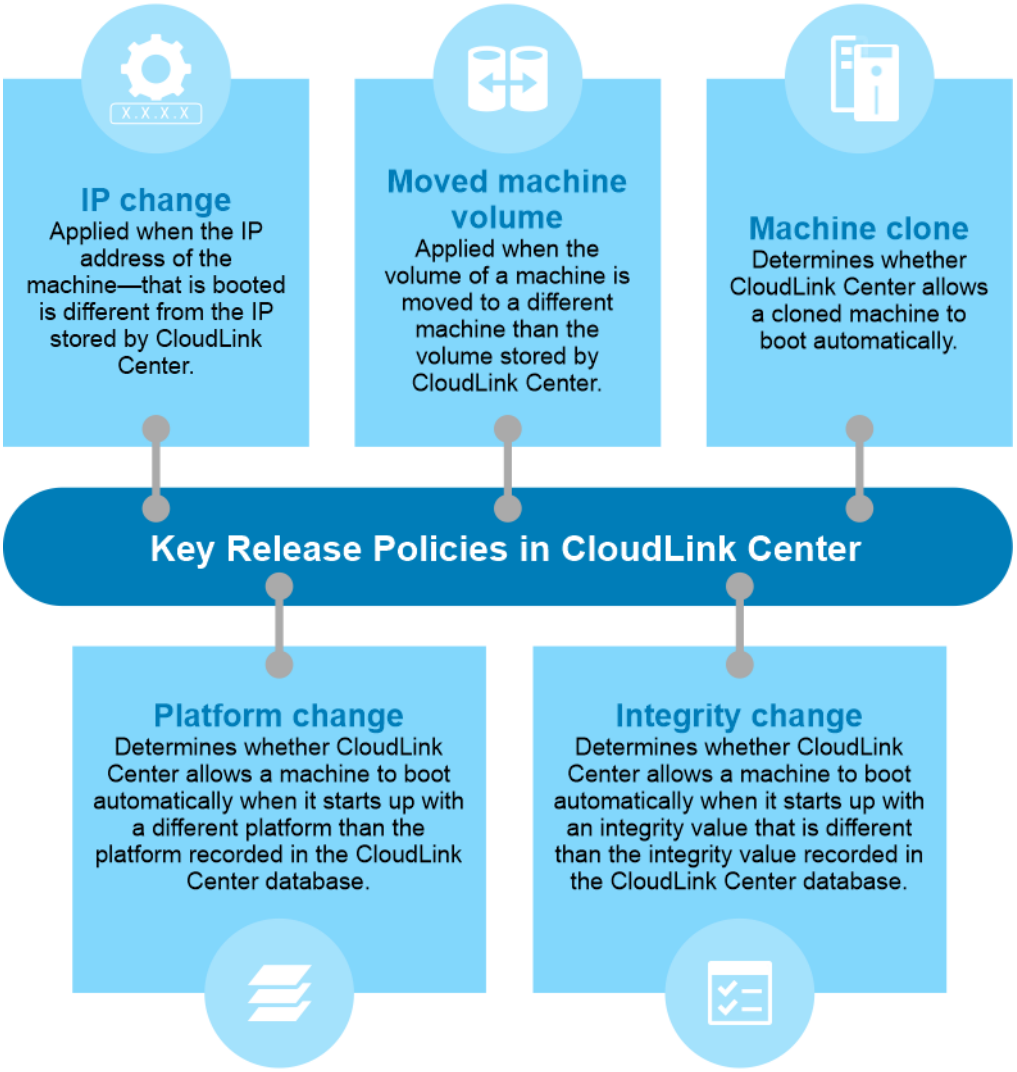
- A machine to boot as part of the prestartup authorization process

- Access to encrypted data volumes or devices.

If a machine does not meet the policies, CloudLink Center puts the machine in the pending state. Manually choose whether to allow the key release.

Key release policies are set for a machine group. For more information, see Manage CloudLink machines by grouping .

The following key release policies are available.



You can change these key release policies. For more information about approved networks, moved volumes or devices, cloned machines, and the pending state, see the *Dell CloudLink Administration Guide*.

Preboot unlock for PowerFlex devices

By changing all the pending policies of a machine group to the **Allow Automatically** mode, unlock the preboot of devices that are connected to PowerFlex machines. This prevents the PowerFlex machine from being placed in the pending state and allows the connected devices to be unlocked.

Manage CloudLink machines by grouping

You can organize machines into groups for administrative or operational purposes. For example, you might group machines for your finance department and apply a volume or a device encryption policy that requires encryption of all boot and data volumes, or devices. Also, you can group machines for your Development and Operations (DevOps) department and apply a volume encryption policy that requires encryption of only boot volumes. Each machine group might have a different administrator.

Each machine must belong to a machine group. A machine is assigned to a machine group during deployment. If you do not specify a group during deployment, the machine is assigned to the in-built machine group named **Default**. After deployment, you can change the machine group of a machine.

All machines in a group use the same:

- Key release policies that determine when CloudLink Center automatically releases keys to a machine. For more information, see CloudLink Key Release Policies.
- Volume encryption policy determines the types of volumes that must be encrypted (boot, data, or both boot and data). Volume encryption policy applies to virtual machines (boot and data volumes).
- Keystore where encryption keys are stored. For more information, see Encryption key location and protection options in CloudLink.
- Only users belonging to a managing role for a machine group can view, change, and perform operations on the machines belonging to it.
- Approved networks, which are network locations that allow automatic startup for machines in a machine group.
- Approved location used to verify that a machine is in the correct place.
- Key lifetime that determines how frequently and at what intervals CloudLink Center updates encryption keys for machines in the group.
- New machine detection policy—If a new machine that has an approved IP address is added to CloudLink Center, you can choose to allow it to automatically register with CloudLink Center (default) or to require manual approval.
- CloudLink Agent upgrade policy, which determines whether CloudLink Agents are upgraded when CloudLink Center is upgraded, or if CloudLink Agents are upgraded individually.

For information about machine groups, volume encryption policy, device encryption policy, managing roles, approved networks, and key lifetimes, see the *Dell CloudLink Administration Guide*.

Encryption key location and protection options in CloudLink

Keystores

The combination of a key location and a key protector. Encryption keys are stored in a key location and are encrypted, or protected, by a key protector.

Key locations

CloudLink Center supports several options for the key location that is used to store encryption keys:

Local Database	An internal key location.
Microsoft Active Directory	An external key location.
Amazon S3	An external key location. You must have an Amazon Web Services (AWS) account to use this location.
S3-compatible bucket	An external S3-compatible key location.
External KMIP Server	In CloudLink Center 7.1.3 and later versions, you can configure an external KMIP server so it can generate and store encryption keys that are required for agent or machine encryption. The key release policies applicable for an agent are applicable for external KMIP servers also. It can be used for encrypting machines, PowerFlex devices, containers, and SED key management. However, the features mentioned earlier are not applicable for Key Management over KMIP, because in this case, CloudLink acts as a KMIP server.

Key protectors

CloudLink Center supports several options for encryption key protectors.

NOTE: The type of available key protector depends on the selected key location.

CloudLink Vault	An internal key protector.
SafeNet LunaSA	An external key protector using a hardware security module (HSM) for protection.
Microsoft Azure or Azure Stack Key Vault	An external key protector using an Azure or Azure Stack Key Vault for protection.
KMIP server	An external key protector using a Key Management Interoperability Protocol (KMIP) server for protection.
Password	The encryption key is protected with a password.

Best practices for saving, backing up, and restoring CloudLink Center machine encryption keys

This topic provides information about the best practices for saving, backing up, and restoring CloudLink Center machine encryption keys.

You are responsible for your encryption keys and for ensuring that the appropriate access control and backup policies and procedures are in place to protect the keys against loss or theft. If your keys become unavailable, you cannot access any data that was encrypted using those keys.

CloudLink Center backups are critical for restoring CloudLink Center. Have a backup of CloudLink Center so that you can deploy a new server and restore CloudLink Center. If you are using the local database, volume encryption keys or device encryption keys are stored in CloudLink Center. Backups are the only method of restoring keys so that you can access encrypted data. For information about VKEKs and volume encryption keys or device encryption keys, see the *Dell CloudLink Administration Guide*.

The following identifies the key protectors that are available for each type of key location.

Key Protector—CloudLink Vault

- Local database key location—Yes
- Microsoft Active Directory key location—No
- Amazon S3 key location—No
- S3-compatible bucket key location—No

Key Protector—SafeNet LunaSA

- Local database key location—Yes
- Microsoft Active Directory key location—No
- Amazon S3 key location—No
- S3-compatible bucket key location—No

Key Protector—Microsoft Azure or Azure Stack Key Vault

- Local database key location—Yes
- Microsoft Active Directory key location—No
- Amazon S3 key location—No
- S3-compatible bucket key location—No

Key Protector—KMIP key manager

- Local database key location—Yes
- Microsoft Active Directory key location—Yes
- Amazon S3 key location—Yes
- S3-compatible bucket key location—Yes

Key Protector—Password

- Local database key location—Yes
- Microsoft Active Directory key location —Yes
- Amazon S3 key location—Yes
- S3-compatible bucket key location—Yes

NOTE: Ensure that you meet all prerequisites for restoring CloudLink Center from backup, otherwise you cannot access encrypted data after restoring from a backup file.

For more information about CloudLink Center backups and restoring from a backup file, see the *Dell CloudLink Administration Guide* .

CloudLink Vault to encrypt CloudLink Center machine data

CloudLink Center includes an encrypted container, CloudLink Vault, which is created during the deployment and configuration of CloudLink. CloudLink Vault:

- Encrypts credentials that are used to access remote resources

  For example, CloudLink Vault stores credentials that are required to access FTP or SFTP servers or external keystores.

- Provides an initial, internal key protector

  You can continue to use this initial CloudLink Vault as the key protector, keystore, or configure a different key protector. When used as the key protector, CloudLink Vault encrypts device or volume key encryption keys (VKEKs). For more information, see the *Dell CloudLink Administration Guide*.

When a CloudLink Center server restarts, it must unlock CloudLink Vault before CloudLink Center can authorize machine operations, ensuring that a stolen copy of CloudLink Vault or the disk on which it is stored does not contain any unprotected secrets or encryption keys.

You can configure CloudLink Vault to open:

- Manually by providing a passcode

  When configuring CloudLink Vault, you specify up to three passcodes. Only one passcode is required to open the vault.

- Automatically by using a server-specific key

NOTE: During CloudLink Center initial configuration, the default vault unlock mode is automatic. You can choose to set the CloudLink Vault to unlock manually.

IP addresses of machines associated with CloudLink

In some circumstances, the IP address of a machine under CloudLink Center management might change, such as when a Dynamic Host Configuration Protocol (DHCP) server assigns IP addresses. When a machine starts up with a changed IP address, CloudLink Center might put the machine in the pending state. Before startup can continue, you must manually accept the machine. For more information about accepting machines in the pending state, see the *Dell CloudLink Administration Guide*.

By default, in Azure and Azure Stack environments, the IP address of a machine under CloudLink Center management might change each time the machine is shut down and restarted. A new IP address is assigned from the same subnet as the previous address. When a machine starts up with a changed IP address, CloudLink Center might put the machine in the pending state. Before startup can continue, you must manually accept the machine.

To avoid manually accepting machines in the pending state because of changed IP addresses, you can change the key release policies for the machine group to allow CloudLink Center to release keys to machines starting up with changed IP addresses. See CloudLink Key Release Policies for more information.

In Azure and Azure Stack environments, to avoid manually accepting machines in the pending state because of changed IP addresses, you can assign static IP addresses to machines.

Manage CloudLink Center clusters

A CloudLink Center cluster provides for high availability if one CloudLink Center server in the cluster becomes unavailable, whether due to planned maintenance or an unexpected issue.

A CloudLink Center cluster consists of up to four CloudLink Center servers, where each server is active always. There is no primary server. The agents can be connected to any server in the cluster.

CloudLink Center replicates configuration information between all servers in a cluster. This replication means that all servers contain the same critical configuration information:



- CloudLink licenses
- Volume encryption policy
- User accounts
- Manual passcodes for unlocking CloudLink Vault
- Actions
- Alarms
- Security events

Data from external resources, such as key locations, key protectors, and key management servers, are not replicated.

For information about creating a CloudLink Center server cluster, see the *Dell CloudLink Administration Guide*.

CloudLink Update menu

After you have configured CloudLink Center, the **Update Menu** is displayed every time you log in using the CloudLink Center.

Update menu options

This topic provide information on Update menu options for Enterprise, PowerFlex, and Microsoft Azure and Azure Stack.

For Enterprise and PowerFlex

- Summary—Displays a summary of CloudLink Center settings.
- Network—Displays CloudLink Center network configuration(s).

- Password—Changes the current password used to log in to the CloudLink Center console.
- Reset Network—Resets the network settings, after which you can reconfigure them. If you select this option, all current network settings are removed.
- Unlock User—Unlocks the secadmin user account.
- Shutdown—Shut down or reboot CloudLink Center.
- Diagnostics—Intended only for use as directed by your Dell Technologies representative.

For Microsoft Azure and Azure Stack

- Summary—Displays a summary of CloudLink Center settings.
- Network—Displays CloudLink Center network configuration(s).
- Unlock User—Unlocks the secadmin user account.
- Shutdown—Shut down or reboot CloudLink Center.
- Diagnostics—Intended only for use as directed by your Dell Technologies representative.

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Related documentation and resources

Other Dell CloudLink documents you may require

You can find the following documents on the Dell Technologies Support website.

- *Dell CloudLink 8.3.x Deployment Guide*
- *Dell CloudLink 8.3.x Deployment Guide*
- *Dell CloudLink 8.3.x Administration Guide*
- *Dell CloudLink 8.3.x Release Notes*
- *Dell CloudLink 8.3.x Upgrade Guide*
- *Dell CloudLink Security Configuration Guide*
- *Dell CloudLink Support Matrix*
- *Dell CloudLink Key Management for VMware vCenter Server Configuration Guide*
- *Dell CloudLink SNMP Reference Guide*

Contact Dell Technologies

Dell Technologies provides several online and telephone-based support and service options. Availability varies by country, region, and product, and some services may not be available in your area. To contact Dell Technologies for sales, technical support, or customer service issues, see Dell support site.

If you do not have an active Internet connection, you can find contact information on your purchase invoice, packing slip, bill, or the product catalog.